Education of Nations

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In memory of Richard A. Easterlin (1926–2024), who asked the right question.

In memory of Jane Goodall (1934–2025), who saw what the juvenile dependency period does.

In memory of Mahbub ul Haq (1934–1998), who knew that people are the real wealth of a nation — and left one step for the rest of us.

Abstract

Education is the necessary and sufficient cause of human development. That claim is what this paper argues, tests, and applies. The one caveat: energy is the thermodynamic substrate every organism runs on — control of fire made the human body and brain possible; agriculture, animal traction, fossil fuels, and now the solar–wind–battery transition each raised the ceiling under which populations could live. Energy is the precondition. Education is where the cross-country variance lives. This paper is about the operative level.

No country has developed without education. No country with sustained educational expansion has failed to develop. The mechanism is biological: humans have the longest juvenile dependency of any species —

nearly two decades of a plastic brain embedded with a dedicated teacher. Extended juvenile dependency is the protocol; education is the most powerful payload ever delivered through it. Educated parents transmit knowledge, norms, and health behaviours to their children, who exercise these as household decisions ~25 years later — the time-to-agency interval set by juvenile dependency itself. A great-grandparent's school completion predicts their great-grandchild's life expectancy three generations later; income's predictive power collapses within one. Income, once education's contribution is removed, predicts nothing. Provision without educated populations produces no durable outcomes. Institutions are built by educated populations, not the reverse. By 2022, 154 countries have crossed both development thresholds (TFR below 3.65, life expectancy above 69.8 — the 1960 United States values) — every one through education. Darwin's theory explained how species differentiate; this paper documents the inverse — the convergence of one species, across every political system and continent, onto the same developmental phenotype through a single biological channel. The evidence is a 185-country panel (1950–2015) tested across three generational depths, with natural experiments that no other development input survives.

Keywords: fundamental cause; parental transmission; residualization; natural experiments; categorical literacy; extended juvenile dependency; energy substrate; development threshold

1. Introduction

Education is fundamental to human well-being. Not important — fundamental. The distinction matters. Important means high on a list. Fundamental means the list does not exist without it.

Education is the necessary and sufficient cause of human development. No country has developed without it. No country with sustained educational expansion has failed to develop — but sufficiency operates across generations, not within individual lifetimes. By "education" this paper means the educated population, not the school system. The school system is the delivery mechanism; the educated adult is the operative unit. A country can rebuild schools within a decade; it cannot rebuild an educated population in less than a generation (Cambodia, Section 5). Educated people are developed — they live longer, have fewer children, make better health decisions, and transmit these gains to their children without requiring income or provision as intermediaries. The mechanism is Parental Transmission (PT): an educated parent transmits knowledge, norms, health behaviours, and the decision to educate the next child. PT carries the gains forward automatically. Population-level development requires the state to extend this to the whole population — not to make education work, but to make uneducated people educated. The state provides reach; the household provides the mechanism.

The evidence is a 185-country panel (WCDE v3, 1950–2015) tested across three generational depths (~25, ~50, ~75 years), with natural experiments that no other development input survives.

Why has this been missed? Every major approach to human development — income-first, institutions-first, provision-first, governance-first — was built by people standing on educational foundations they could not see. Adam Smith (1776) observed the pin factory and built a theory of wealth from the division of labour. He never asked where the educated workforce came from. Scotland's Education Act of 1696 had been producing educated Scots for eighty years before Smith was born; he took educated labour as given — the same structural blindness that makes education invisible to educated populations today (Sections 2.4 and 4.3).

Japan, with negligible fossil fuels, industrialised within a generation of the

Meiji education reforms; Nigeria, sitting on one of the world's largest oil reserves, did not. Every major development framework has made the same structural move: mistaking the products of education for its causes. The policy consequence is dispersion: resources spread across interventions, with education funded as one line item rather than recognised as the mechanism that makes the others work.

Smith's blindspot had a twin. The same framework that took educated labour as given also took abundant coal as given. Energy is the substrate every organism runs on; humans are the variant that layered cultural transmission onto it. The argument in this paper names both layers: energy as the substrate shared with all life, education as the species-specific mechanism through which the human variant becomes developmental at population scale. The empirical work is about the operative layer because that is where the cross-country variance lives. The substrate runs under every case the paper analyses and every prediction it makes (Section 2.1).

Others have seen parts of this, from two directions. In development economics, Haq & Sen created the Human Development Index (UNDP 1990), breaking the identification of development with income. Sen's capabilities framework (1999) showed that what matters is what people can do and be, not what they earn — and he saw the right anomaly: Kerala, Sri Lanka, China, with welfare outcomes far above what their incomes predicted. Easterlin (1981) identified mass schooling as the binding constraint. Lutz & Kebede (2018) showed the income–mortality relationship is tighter when education replaces income on the x-axis: what looked like "wealth buys health" was education proxying through income all along.

In evolutionary biology, Konner (2010) documented the longest juvenile dependency period in any species — roughly 18 years of dedicated learning embedded with a dedicated teacher. Hrdy (2009) showed that a dependency of this length was only viable because humans evolved as cooperative breeders: mothers, fathers, grandparents, older siblings, and alloparents shared the burden of provisioning children that no pair of parents could carry alone. The household this paper measures is the one Hrdy identifies. Boyd & Richerson (1985) formalised cultural transmission through parent-to-child channels as the mechanism by which human populations accumulate knowledge across

generations. Hawkes et al. (1998) showed that human post-reproductive lifespans — unique among primates — evolved to support grandchild survival, making longevity and fertility two expressions of the same biological architecture.

These three findings form a chain. Hamilton (1964) formalised inclusive fitness: behaviour that raises the reproductive success of kin is selected even when it lowers the actor's own. Boyd & Richerson (1985) extended the logic from genes to culture: knowledge, norms, and the decision to educate are themselves heritable and subject to the same selective pressures, with the parent-to-child channel the highest-fidelity route. Hawkes et al. (1998) closed the biology: human post-reproductive longevity is itself a grandchild-survival trait, which is why fertility and life expectancy move together as one developmental signature. Parental Transmission is inclusive fitness operating through cultural rather than genetic inheritance — kin-directed investment in descendants, carried by the extended juvenile dependency period and supported by the post-reproductive longevity that evolved alongside it. This paper applies evolutionary biology to explain human development. The empirical domain is human societies over the past seventy years.

These traditions — one asking what development is, the other asking what humans are — converge in this paper's central claim: that education is the cause, income and provision are downstream, and the mechanism operates through the biological channel of parental transmission across generations. This paper states it and tests it.

Section 2 shows that extended juvenile dependency makes education the only cultural payload that transmits automatically across generations. Section 3 defines development through the two outcomes — life expectancy and fertility — that summarise every other development metric. Section 4 shows that income, once education's contribution is removed, predicts nothing; that education's signal survives across four generations while no competitor survives one; and that no educated population has experienced famine without external force. Section 5 traces the mechanism through eight cases in which one variable held while others varied. Section 6 states the policy consequence: with lower secondary completion as the singular priority, Uganda reaches the development threshold by 2050.

2. What Humans Are

2.1 The Energy Substrate

Before what humans do with education, what all life does with energy.

Every organism is an energy-extraction system. Life, at its most abstract, is the local reversal of entropy paid for by the capture and dissipation of environmental energy (Smil 2017). Humans are a variant of this general pattern, with a particular history. The control of fire allowed cooking, which raised the net caloric yield of food enough to support the metabolically expensive human brain (Wrangham 2009). The human body and brain are products of that substrate. We did not first become cognitively modern and then learn to cook.

Every subsequent enlargement of the human population was underwritten by an enlargement of energy capture: agriculture displaced foraging, animal traction and wind displaced human muscle, fossil fuels displaced all of it (Wrigley 2010; Smil 2017). Each step released the population ceiling of the regime preceding it. Mass education in the modern sense became materially possible because the fossil-fuel economy produced the surplus to sustain schools, teachers, and the extraction of children from productive labour for the duration of their juvenile dependency. Scotland in 1696 could afford Knox's parish schools. Most of the world in 1696 could not.

The paper's biological mechanism sits on this substrate without being reducible to it. Energy is shared with all life; extended juvenile dependency, parental transmission, and the formal schooling that produces categorical literacy are not. Two populations with the same per-capita energy — one that educates, one that does not — produce different development outcomes. Energy is the precondition under which the mechanism operates; it is not the mechanism. The rest of this section is about the species-specific layer.

2.2 The Generational Transmission Mechanism

Educated parents pass what they know to their children, who pass it to theirs. This is the mechanism.

PT (Parental Transmission) is cultural transmission through the household. The channel is biological — the extended juvenile dependency period co-resident with grandparents, older siblings, and extended kin, the cooperative-breeding unit Hrdy (2009) identifies — but what travels through it is cultural: norms, aspirations, health behaviours, and the decision to educate the next child. Of these, the paper measures one: education completion, which is both the operative item the household transmits and the proxy for everything else. PT runs on self-interest: household members invest in their own children without requiring ideology, altruism, or institutional commitment to sustain the behaviour. It transmits across generations: the grandchild's attainment reflects the grandparent's through two sequential cycles (~50 years), the great-grandchild's through three (~75 years).

The operative transmission is the decision to educate the next child. A parent who completed school sends their child to school — with expectation, monitoring, and reinforcement. A parent who did not, does not — or does so without any of these. Completion transmits as a norm. The completion threshold is the gate; what happens after the gate is dose-response (Section 4.3). PT explains persistence — why educational gains never depreciate. But persistence alone does not produce development outcomes. What matters is what education creates in the individual: *categorical literacy* — the reorganisation of the brain that literacy and numeracy physically produce (Dehaene et al. 2010; Section 4.3), giving the capacity for planning, health-oriented decision-making, and information processing that does not develop without schooling. This capacity is exercised through household decisions: fertility, health, investment, and children's schooling. The ~25-year lag between educational investment and development outcomes is the interval for educated children to become adults exercising these decisions. Taiwan's development crossing arrived ~20 years after educational expansion began: one generation. Kerala's arrived at ~65 years: three generations of slow accumulation before enough of the population was educated for household decisions to produce population-level development outcomes. A woman completing lower secondary at age 15–18 has children reaching school age 20–30 years later.

Educational expansion has two parts. PT provides durability: it transmits what parents already have. State investment provides reach: it brings first-generation learners into the system and lifts the level beyond what parents

alone would transmit. Where the state invests, both channels operate together — parents transmit what they have while the state reaches children whose parents have nothing (Section 4.5). Where the state does not invest, PT alone preserves what the parent already has — nearly perfectly, but nothing more.

PT is the mechanism in every case; what varies is the state's relationship to it. **No priority**: the state does not invest. Pre-industrial, this produced stasis; post-industrial, educated minorities with surplus and widened identity (Section 2.3) slowly expanded access through religious and community channels, but never singular commitment (Section 6). **Competing priority**: the state builds schools, but education competes with health, nutrition, security, and governance. **Singular priority**: the state makes education the unconditional focus and builds ahead of need. The cases (Section 5) develop all three regimes.

This architecture explains three properties. Non-depreciation: education persists for life because what survives is not content but the cognitive reorganisation that content exposure produced (Section 4.3). Amplification: when the state builds schools, it loads additional knowledge onto the same channel — the $\beta_g > 1$ result, where β_g is the ratio of the child generation's education gain to the parental baseline (Section 4.5). Catastrophic reset: because the knowledge lives in the teacher, not the institution, destroying the educated population leaves the next generation's plastic brains with nothing to absorb. Cambodia (Section 5) is this test: schools rebuilt within a decade, progress stalled for a generation. Knox did not invent parental transmission; he institutionalised it, channelling formal schooling through a mechanism as old as the species.

The human baseline — the state before formal education reaches a population — is high fertility, high child mortality, low life expectancy, hunger, and disease. This is not a description of poverty. It is a description of the species in the absence of the one intervention that changes all of them simultaneously. Hunger and insecurity are part of the baseline, not preconditions that must be resolved before education can begin. Every successful educational expansion in the historical record was made through poor nutrition and insecurity, not after resolving them (Section 5).

The pathway runs: education restructures cognition; restructured cognition restructures intent (family size becomes a choice, child survival becomes an expectation); the fertility transition follows; fewer children releases resources per child; those resources amplify the next generation's education through PT; the cycle repeats. Each pass raises the baseline.

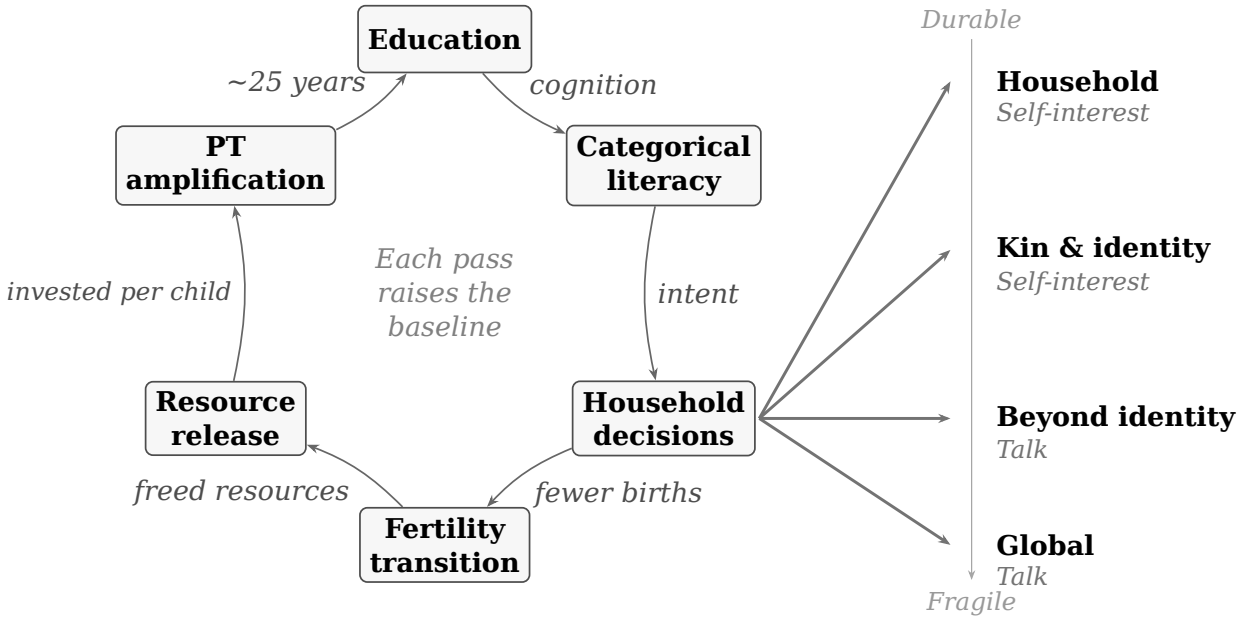

Figure 1: The generational transmission loop and four radii of educational effect. The loop traces the causal pathway from education through household decisions back to the next generation's education (~25 years). The four radii show how education's effects extend beyond the household with decreasing durability: self-interest compounds; talk requires continuous external commitment.

2.3 From Action to Talk: How Education Reaches Beyond the Household

Education's effects start in the household but do not stay there. The four radii shown in Figure 1 extend education's effects beyond the generational loop, with decreasing durability as the radius widens.

The PT mechanism operates through action — the household decisions of educated adults. But education's effects extend beyond the household, and the nature of that extension changes as the radius widens. Two things widen it simultaneously: education expands the boundary of who counts as "my people" — coalitional psychology (Tooby & Cosmides 2010) allows flexible group

boundaries that education stretches from kin to nation — and it releases surplus, as educated parents have fewer children and nearly all survive, freeing time, money, and attention. The arc widens (who you invest in) and the constraint loosens (what you can invest).

The result is four radii of effect, each weaker than the last:

1. **Household.** PT is action: the parent raises the child in front of them, and the 18-year biological channel guarantees delivery. Runs on self-interest.
2. **Kin and identity.** Investment is still concrete and compounding. Runs on self-interest extended by expanded identity.
3. **Beyond identity.** Education becomes talk — advocacy, policy campaigns, diluted across every good cause simultaneously. This is where the competing-priority pathway (Section 2.2) originates.
4. **Global.** External provision — vaccines, aid, health delivery — funded by the surplus of already-developed populations. Talk, diluted further across borders and competing claims.

The axis that predicts durability is motivational: PT compounds because it runs on self-interest. Every non-education intervention requires someone to keep serving past the point where self-interest would redirect them. Revolutionary commitment, donor altruism, state fiscal discipline: all are fragile because all run against the grain of normal human motivation. Education runs with it. Non-education intervention buys time; educational investment changes trajectory.

2.4 Why Education Is Invisible

The mechanism is invisible to the populations it produces. Three claims make this precise.

(i) Education leaves no physical signature. An educated person is physically indistinguishable from an uneducated one. Same body, same biological drives, same appearance. Wealth builds visible infrastructure; disease marks the body; education rewires cognition and leaves no trace on the surface. The only observable signal is behavioural change — and we misread it. We

see smaller families and call it wealth flows, or contraceptive access, or patriarchy relaxing, or telenovelas on television. We see longer lives and call it health systems. We see rising productivity and call it economic growth. Every discipline theorises the downstream behaviour without tracing it back to the cognitive reorganisation that produced it. Education has no physical signature, so the signature it does leave — changed decisions, across every domain, for the rest of the person's life — is assigned to whatever domain notices it first.

(ii) No observer can rank education as fundamental. Educated people rank education as important — one of many important things, alongside health, nutrition, governance, and security. They cannot categorise it as *fundamental* because that would require seeing their own cognition from outside. Uneducated people cannot value it at all: the cost is immediate, the return is abstract, delayed by a generation, and has no precedent in the household's experience. The people who could identify education as fundamental cannot, because they are inside it; the people who would benefit most cannot, because they are outside it. Subjective wellbeing confirms the blindspot: across 147 countries, self-reported happiness tracks income, not education — people feel richer, not more educated, so they attribute progress to wealth.[1]

This applies to the frameworks through which this paper will be evaluated. Development as multivariate, education as one input among many — these are themselves products of the mechanism the paper describes. The capacity to rank, compare, and weigh competing inputs is itself a product of education. The training that makes "education is sufficient" sound like overreach assumes a multivariate world. The evidence will show that the other variables are either products of education or irrelevant once education is present.

The clearest contemporary demonstration is the United Nations Sustainable Development Goals (UN 2015). The SDG framework was designed by the most credentialed development institutions on earth, with full access to the evidence this paper cites — including Haq and Sen's own Human Development Index, Easterlin's identification of mass schooling as the binding constraint, and the Preston Curve literature showing education proxying through income. The result: SDG 4, Quality Education, listed fourth among seventeen

[1] See `scripts/happiness_education.py` and `scripts/figures/fig_happiness_edu_gdp.py`.

parallel goals — alongside SDG 1 (No Poverty), SDG 2 (Zero Hunger), SDG 3 (Good Health), SDG 5 (Gender Equality), SDG 8 (Decent Work), and SDG 10 (Reduced Inequalities). Every one of these is a downstream product of education. Income is what educated populations generate — stripped of education's contribution it predicts nothing (Section 4.7). Hunger is what educated households escape — not through provision but through the fertility transition that releases resources per child, the planning horizon that converts subsistence into surplus, and the household-level cognitive and behavioural capacity that prevents food shocks from becoming famines: of 21 famines since 1950, 19 occurred where lower secondary completion was below 50%; of 17 comparable shocks in high-education populations, none produced famine (Section 4.10). Gender equality is what the fertility transition produces first — the shift from biological fate to conscious reproductive choice that occurs at primary education, before labour force participation, before political voice, before any of the dimensions SDG 5 measures independently.

The SDG framework does not reflect ignorance of the evidence. It reflects the mechanism: the educated people who designed it experienced income, food security, health literacy, and institutional capacity as the normal baseline from which you design frameworks — not as products of education that the uneducated world does not yet have. Education at number four, competing for budget allocations against hunger programs that feed children today, poverty interventions with visible short-term outcomes, and health delivery systems that save lives this year, is not an oversight. It is what the competing-priority regime looks like when institutionalised at the global scale by people who cannot see they are inside it. No previous development framework has been built by people who could see outside it. This one was not either.

(iii) Historical breakthroughs bypassed the frame entirely. Inside the development frame, education competes with health, nutrition, governance, and security in a cost-benefit comparison — and it always loses, because its returns are generational while the others' are immediate and visible. A child dying of diarrhoea today is an emergency; a child not sent to school today is no one's emergency — until her child arrives as one in 2050.

Every historical breakthrough bypassed this comparison: Knox, Meiji Japan, Korea, and Cuba made education non-negotiable for reasons that preceded

any cost-benefit calculation — salvation, existential military threat, Cold War survival, revolutionary ideology. The policy consequence — what it takes to replicate a bypass from evidence rather than contingency — is Section 6.

2.5 The Argument Applied to Itself

The argument of this paper applies to its own production, and to the production of every argument, discovery, and specification in the history of ideas. Its author received categorical literacy through formal schooling. The biological components of the mechanism — Konner on juvenile dependency, Boyd and Richerson on cultural transmission, Hawkes on post-reproductive lifespan, Dehaene on literacy's neural reorganisation — and the development components — Haq, Sen, Easterlin, Lutz and Kebede — arrived through the same transmission channel the paper describes. The synthesis is that channel operating on itself: an instance of what educated minds do with accumulated payload, no different in kind from any other output of the mechanism.

Every foundational specification is produced this way. Smith wrote from within an educated Scotland whose parish schools had been building literate labour for a century before him; he took the literate workforce as given and theorised the division of labour it enabled. Darwin wrote from within the accumulated tradition of Linnaean taxonomy, Lyellian geology, and Malthusian population dynamics, each transmitted to him through formal schooling at Edinburgh and Cambridge and through the cultural networks of Victorian science. Wallace reached the same synthesis independently because the tradition was ready. Shannon wrote from within the mathematical-engineering tradition of Nyquist, Hartley, Wiener, and Turing. Mendeleev assembled Döbereiner, Newlands, and Meyer into the periodic table; Meyer reached a parallel arrangement the same year. In each case the specifier is a node in a transmission chain that has reached the threshold where the chain's own mechanism can be articulated from within.

Traditions specify their own mechanisms at readiness-moments. The specific author is contingent; the readiness is not. Parallel synthesis — Leibniz and Newton, Wallace and Darwin, Meyer and Mendeleev — is the usual signature of a tradition arriving at its own self-description. What matters is that the accumulated components reach the point where synthesis becomes possible,

not who performs the synthesis.

This closes the invisibility argument of the previous section. The capacity to see the mechanism from inside is produced by the mechanism itself, operating across enough generations of accumulated knowledge that the channel becomes describable in its own terms. Education is invisible at shallow depths of accumulated understanding and specifiable at sufficient ones. The present synthesis is the tradition Haq and Sen opened in 1990 reaching the depth at which its own operative channel can be named. The paper is the channel speaking about itself — late, contingent, and no more exterior to the phenomenon than any other educated output ever has been.

2.6 Demographic Structure and the Fertility Transition

Educated women have fewer children. The decline is continuous, steepest at the bottom, and requires no threshold to trigger.

Fertility declines continuously as female secondary completion rises. The evolutionary mechanism is a shift in reproductive strategy, not an override of the reproductive drive: in high-mortality environments, the optimal strategy is quantity — have many children because some will die. Education changes both inputs to this calculus. Child survival rises (fewer births needed to achieve the same number of surviving children), and the planning horizon extends — a literate parent can see that investing more in fewer children produces better outcomes than hedging with numbers. The drive to maximise offspring success is unchanged; what changes is the strategy that serves it.

The steepest fertility decline occurs at primary education (Section 4.7), where women are barely literate. What primary education gives is capacity over preferences that already exist — the capacity to read a contraceptive instruction, understand a health protocol, act on a desire not to be pregnant continuously. Yet the demographic headwinds are also steepest here: where total fertility rates are typically above 4, population growth is rapid, and the gains from each educated cohort are diluted across a larger and faster-growing next generation. As completion rises, fertility falls, child survival improves, and household resources per child increase. PT is a household mechanism. States can create conditions for these decisions; they cannot make them.

Human biology enforces the scale of these decisions. A woman who begins bearing children at fifteen and has seven will still be pregnant or nursing at thirty-five — and the last child, born then, still needs daily care when she is in her mid-forties. High fertility is not merely correlated with constrained lives; it is the constraint.

Even in populations where life expectancy at birth is low, women who survive to childbearing age typically live into their fifties or beyond; the average is dragged down by infant and child death, not by adults dying at forty. But survival does not mean freedom. The grandmother hypothesis in evolutionary biology holds that human females live decades past menopause precisely because post-reproductive women raised grandchildren's survival odds. The long post-reproductive lifespan is itself part of the transmission architecture. Life expectancy and fertility are not independent metrics that happen to correlate — they are two readouts of the same biological architecture. The species evolved long post-reproductive lives because it evolved extended juvenile dependency because it evolved cultural transmission. Together they are the biological signature of the human developmental niche.

Fertility decline is not a proxy for women's empowerment — it is the first expression of it. Education gives capacity over preferences that already exist: contraceptive availability alone does not produce the shift; education does. The further dimensions of empowerment that the literature measures — labour force participation, household bargaining power, political voice, economic independence — follow after fertility falls, as smaller families free time, resources, and attention. Education produces the first move; the rest cascades from it.

2.7 Predictions

These predictions follow from the biology, not from the data — they are stated before any empirical test is run. The predictions were specified first; Section 4 is the formal check. If the mechanism described above is correct — if extended juvenile dependency creates a biological channel for cultural transmission, and formal education is the most powerful payload ever delivered through that channel — then six empirical patterns must hold.

1. **Generational timing.** Effects on development outcomes should appear

at ~25 years — the time-to-agency interval set by juvenile dependency — not at 10 or 15 years.

2. **Multi-generational persistence.** Education's predictive power should decay slowly across generational depths (50, 75, 100 years), because PT transmits through the biological channel from parent to child to grandchild. Income, which has no equivalent transmission mechanism, should collapse within one generation.

3. **Asymmetric disruption.** Destroying the educated population should reset generational progress, because knowledge lives in the carrier, not the institution. Destroying institutions, income, or infrastructure while leaving the educated population intact should not.

4. **Income independence.** Income stripped of education's contribution should have no independent predictive power for development outcomes, because the causal path runs through education (education $\rightarrow$ cognitive capacity $\rightarrow$ household decisions $\rightarrow$ outcomes).

5. **Universality.** The pattern should hold across political systems, cultures, geographies, and colonial histories, because extended juvenile dependency is a species-level trait, not an institutional product.

6. **Amplification at low baselines.** Each generation's attainment should exceed the parental baseline by the most where that baseline is lowest, because state reach adds first-generation learners on top of what PT transmits; at high baselines the gap should close as ceiling compression sets in. Formally: the generational amplification ratio $\beta_g > 1$ at low baselines, $\beta_g \rightarrow 0$ at high ones.

Section 4 tests all six.

3. Defining Development

I define development as people living long lives and women having fewer children. I measure it concretely: a country is developed when it has simultaneously achieved a **total fertility rate (TFR) below 3.65** and **life expectancy > 69.8 years** — the 1960 United States values (World Bank WDI).

These two measures are chosen because they are intrinsically valuable — not as proxies for something else, but as the things themselves. The UN's Human Development Index (UNDP 1990) already treats health and education as intrinsically valuable alongside income. This definition takes that logic to its conclusion: it retains the two that are ends in themselves and drops the one that is not.

Life expectancy is the most fundamental measure of human welfare. Every other outcome — income, freedom, knowledge, dignity — presupposes being alive. A country where people die at 45 has failed on the most basic term, regardless of its GDP or institutional quality. Life expectancy is not one metric among many; it is the precondition for all others.

The number of children a woman bears defines the shape of her entire adult life in ways that no other variable does. Seven children means twenty years of continuous pregnancy and nursing, followed by a further decade of child-rearing before the last child is self-sufficient (Section 2.6). The difference between seven children and two is not incremental; it is the difference between a life fully absorbed by reproduction and a life that permits education, work, civic participation, or rest.

These are also the two most direct population-level expressions of whether the cognitive transformation produced by education has occurred. Life expectancy tracks whether people have the planning horizon, health knowledge, and behavioural capacity to sustain their own survival. TFR tracks whether women have moved from biological fate to conscious reproductive choice — precisely the cognitive autonomy that underlies every other dimension of human flourishing.

Every competing metric of development is either a component of life expectancy and fertility — child mortality is a component of LE, nutrition is a cause of LE variation, sanitation prevents child death that shows up in LE — or a consequence of the cognitive transformation that moves them (institutional quality, income, democracy). No input can outrank the summary of itself. No consequence can outrank its cause. LE and TFR are not the best items on a list of development metrics — they are what the list is measuring.

No country considered developed — by any definition, under any framework

— has life expectancy below 69.8 or TFR above 3.65. Even Sen's capabilities — health, bodily integrity, political participation, practical reason — all presuppose being alive and having cognitive autonomy over one's reproductive life. Go through the list: political participation requires forming preferences and evaluating evidence; bodily health requires understanding what health means and acting on it; practical reason requires the planning horizon that education produces. Every capability on Sen's list, examined for its generative source, has educated cognition as its enabling precondition. On his own framework, LE and TFR are not two capabilities among many; they are the enabling conditions for the capability to have capabilities.

They require neither high income nor a particular type of government to achieve. Crossing dates under three threshold specifications shift by 10–30 years but the ordering holds (Table A6). Countries with high expansion rates cross any threshold within 14–16 years. GDP is excluded: measuring development through income would assume the answer. For readers who include income in their definition: education predicts future GDP per capita as strongly as it predicts LE and TFR (Table 3). The definition chosen here is the conservative one — it strengthens the alternative's position by removing from the outcome the one variable where income might have independent standing.

Why the 1960 USA? It is the natural baseline: 1960 marks the start of mass decolonization, and the USA was the unambiguous global hegemon. A country that crosses both thresholds has surpassed the richest nation on earth on the metrics that matter for human welfare. The USA in 1960 was mid-transition — its TFR of 3.65 was the baby boom peak, and it did not complete its own fertility transition until 1972. A TFR below 3.65 is itself an achievement: in 1960, only the developed world — Europe, Japan, and a few settler states, roughly a quarter of humanity — had fertility that low. The remaining three-quarters of humanity averaged a TFR above 6 — the human baseline.

The countries that had already crossed both thresholds by 1960 were those with the longest prior histories of mass education: Scandinavia, the Netherlands, France, Switzerland, the United Kingdom, Australia, and New Zealand. Japan — despite a TFR of 2.0 — had life expectancy of 67.7, just below the threshold; it crossed in 1964. By 2022, 154 countries have crossed both thresholds, representing 80% of the world's population. Figure 2 traces this

accumulation over time.

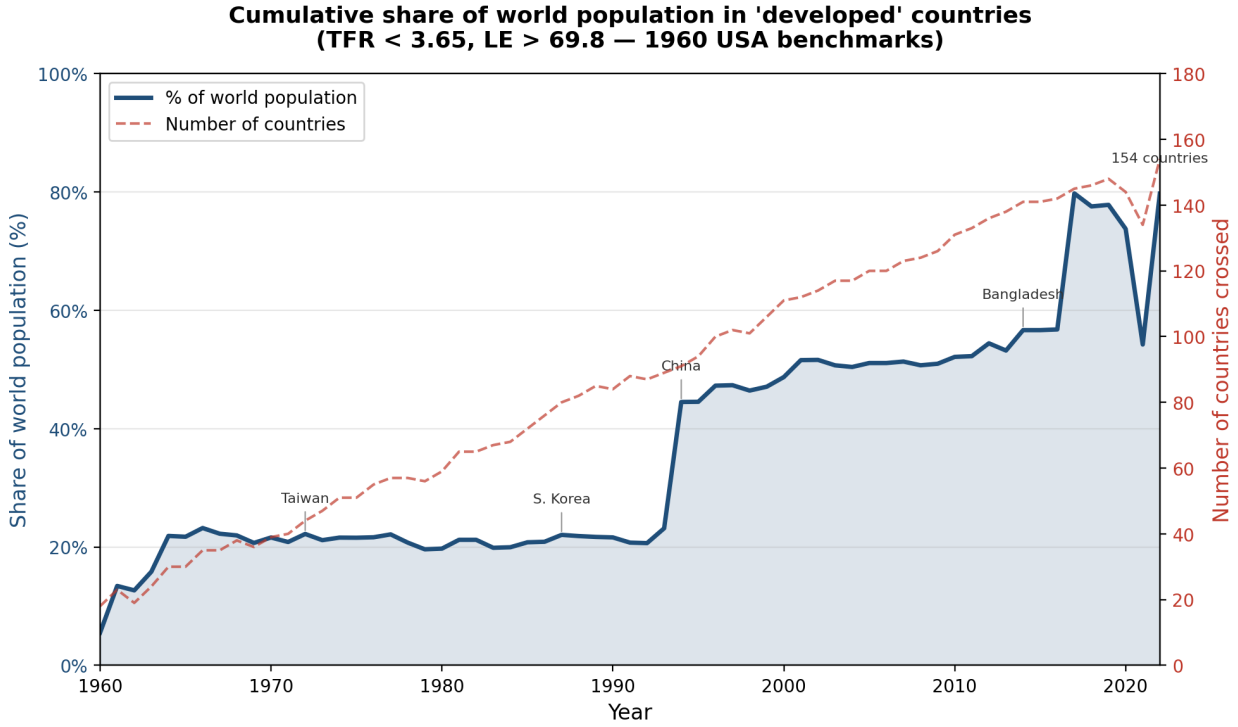

Figure 2: Cumulative share of world population crossing both development thresholds (TFR < 3.65, LE > 69.8), 1960–2022. Slow climb from 13% (1961) to ~20% (1993); vertical jump in 1994 when China crosses; 50% by 2001; 80% by 2017–2022. *(Source: World Bank WDI, WCDE v3. See* `scripts/figures/cumulative_developed.py`*)*

4. The Evidence

Education predicts every development outcome one generation forward; income, once education's contribution is removed, predicts none of them. These results are not found by searching for a pattern — they are the six predictions of the biological mechanism (Section 2.7), tested against 185 countries over 65 years. All six hold.

The statistical tests below formalise what the cases in Section 5 demonstrate historically.

No single table in this section establishes causation on its own. Four tests rule out four alternative stories:

- **Forward lag** (Table 3) — rules out reverse causality.

- **Residualization** (Table 6) — rules out GDP-as-independent-cause.
- **Shock test** (Table 10) — rules out institutions/income as sufficient.
- **Famine test** (§4.10) — rules out provision as necessary.

Each test admits an alternative story in isolation. The causal case is that no alternative explanation survives all of them simultaneously.

4.1 Temporal Structure and Mechanism Testing

Education predicts income one generation forward (Table 3). Because education produces income, GDP sits on the causal path (education → GDP → life expectancy, fertility). Controlling for GDP strips out part of education's effect and biases the result against the mechanism (Pearl 2009; Angrist & Pischke 2009, ch. 3). The confusion arises from treating income as a cause when it is a product of education. Regressing outcomes on income compares education's product against the education that produced it — with the advantage given to the product.

The identification rests on three structures instead. First, temporal ordering: parental cohort precedes child cohort by biological necessity, ruling out reverse causality at the individual level. Second, the lag-decay shape (Figure 4; Section 4.6): education and income produce structurally different decay profiles that autocorrelation cannot explain. Third, and most directly, a set of natural experiments that test Prediction 3 — cases where parental education was exogenously disrupted or where other development inputs were removed while education persisted — produce outcomes that pure autocorrelation cannot predict.

The set includes: Cambodia (population destruction → generational shadow); Sri Lanka (civil war, PT unbroken, development on schedule); Myanmar (73 years of military rule, outcomes tracked education); China (barefoot doctors dismantled, LE continued rising); Korea vs Philippines (identical colonial base, divergent state commitment); and the Asian Financial Crisis (income wiped, education untouched). The cases are developed in Section 5.

The symmetry objection. If GDP mediates education's effect, controlling for GDP biases against education — but if education partially mediates GDP's

effect, controlling for education biases against GDP. Three empirical results break the symmetry: (i) education's predictive advantage over GDP widens to 3.3× among countries still building their education systems (Section 4.5), while GDP's predictive power is flat across all cutoffs; (ii) education's predictive power persists across multiple generations while income's collapses within one (Figure 4; Section 4.6); and (iii) natural experiments that remove income while preserving education show development continuing on schedule (Section 5). Autocorrelation and reverse causality cannot produce this pattern.

Individual-level designs converge on the same direction.

Duflo's (2001) analysis of Indonesia's INPRES school construction programme — the cleanest exogenous education shock in the literature — finds that treated cohorts completed more schooling and earned higher wages, with intergenerational follow-ups showing the effect carries to their children. These designs identify at a different level of analysis (individual cohorts rather than population generations) and cannot test multi-generational persistence or residualization against GDP, but they are consistent with the mechanism specified here.

4.2 Data

The analysis covers 185 countries from 1950 to 2015, with education measured as the share of 20–24-year-olds who finished lower secondary school.

Education data are from WCDE v3 (Lutz et al. 2021): lower secondary completion rates for the 20–24 age cohort, both sexes, which reflects completed education rather than enrolment. All education percentages in this paper refer to this measure unless otherwise noted. Coverage: 185 countries, 1950–2015 at five-year intervals. The core generational panel uses 1975–2015 (1,665 country-years); the residualization analysis (Section 4.7) extends to T=1960–1990 with education interpolated to annual values for precise threshold identification. Long-run analysis uses a 28-country subsample extending back to 1900, selected for data availability; the generational coefficient varies systematically with baseline education level (Section 4.5). GDP data: World Bank, constant 2017 USD, per capita, log-transformed. Life expectancy, TFR, and under-5 mortality: World Bank WDI. Parental education: each country's lower

secondary completion lagged 25 years (T−25).

4.3 Completion as the Operative Variable

This paper measures whether people finished school, not how well they scored on tests. Completion is the right measure for a generational mechanism.

Formal education — even at the lowest quality — represents a categorical discontinuity from oral cultural transmission. The neuroscience is unambiguous: literacy physically restructures the brain's visual processing pathways (Dehaene et al. 2010), and numeracy recruits parietal circuits unavailable to innumerate populations (Gordon 2004; Frank et al. 2008). These are not marginal improvements on existing capacities — they are new cognitive technologies — *categorical literacy* — that do not develop without schooling, regardless of intelligence or environment. An educated person knows bacteria cause disease; an uneducated person does not — there is no halfway position between germ theory and miasma.

Why exposure, not learning, is the operative variable. The first-generation learner undergoes a categorical change by being in school — not by scoring well on tests. What formal schooling provides is not a specific curriculum but sustained exposure to literate culture: structured time, abstract categorisation, written language, numeracy, adult authority organised around knowledge transmission. The longer the child remains inside this environment, the deeper the cognitive reorganisation. Primary education gives capacity over reproduction — the ability to read a contraceptive instruction, understand a health protocol, act on desires that already exist. Secondary gives capacity over health navigation, economic participation, civic engagement. Tertiary gives capacity over complex systems. Each level is not a separate input; it is more time inside the categorical change. The dose is duration of exposure, not fidelity of instruction.

This is why completion predicts development outcomes and test scores do not. Hanushek & Woessmann (2008, 2015) measure what the child learned — the payload's fidelity. The mechanism operates on the exposure itself. A child who completed nine years of poor-quality schooling and scored badly on every assessment has still spent nine years inside an environment that reorganised

their cognition categorically relative to a parent who never entered the building. The test score measures how well the payload was delivered. Completion measures whether the child was inside the channel long enough for the reorganisation to consolidate. Development outcomes track the second, not the first.

Why content does not matter. Tests measure breadth of content retention at a point in time. Content is the dimension that disappears. A software engineer does not remember 10th-grade biology. A doctor does not remember trigonometry. Decades later, the specific material is gone. If the mechanism operated through content — through what the student learned — then education would depreciate as content is forgotten. It does not. No generation in any country has completed school at a lower rate than its parents, and the life-course returns to education show no decay with age. The gains persist because they do not run through content. They run through something that content exposure produced but that survives the content's disappearance.

What survives: cognitive reorganisation. What survives is the cumulative cognitive reorganisation. Years of sustained exposure to biology, chemistry, history, mathematics, and literature do not produce adults who remember all five. They produce adults who spent years inside five different formal knowledge systems — each with its own mode of abstraction, its own vocabulary, its own relationship between evidence and conclusion. The breadth of the curriculum is not an attempt to produce polymaths. It is sustained duration of exposure across multiple modes of formal reasoning, each contributing to the same cumulative reorganisation. Tests measure whether the student retained the content of each subject. The mechanism does not operate through retention. It operates through the years the student spent inside the reorganising environment. The content is the vehicle; the reorganisation is the payload. The vehicle is disposable. The payload is permanent.

This connects the non-depreciation result to the completion-vs-quality debate. Tests predict that education should depreciate — content is forgotten, scores would decline if re-tested, specific knowledge fades. Education does not depreciate. Therefore whatever persists is not what tests measure. Completion captures it. Test scores do not.

The generational gate. The distinction sharpens at the generational gate. The educated parent's role in PT is not to teach their child what they learned in school. It is to hold the expectation that their child belongs in school — to see completion as normal, to monitor attendance, to assume that the educated world is where their child will live. This expectation is the gate. An uneducated parent cannot open it, because they have no experience of the world on the other side. An educated parent opens it automatically — through self-interest, not ideology — because school is where they came from and where their child belongs. The gate does not require the parent to be a good teacher. It requires the parent to have been a student.

This is also why the grandparent effect is strongest at low baselines. Where education is scarce, the grandparent is a second educated person in the household — and, in a village where education is rare, a visible exemplar to the wider community. At high baselines, the entire society holds this expectation and the grandparent's independent contribution disappears — exactly the pattern the data show (grandparent β significant below 50% parental completion, zero above it).

From reorganisation to specialisation. The reorganised individual naturally specialises. Categorical literacy — the capacity for systematic learning, abstract reasoning, structured problem-solving — is not an end state. It is the precondition for further skill acquisition. The educated individual can absorb trade skills through apprenticeship, on-the-job training through employment, formal knowledge through college, deep expertise through a PhD. The specific pathway does not matter. What matters is that each pathway requires the cognitive infrastructure that formal schooling built. The uneducated individual is not excluded from specialisation by lack of opportunity alone. They are excluded because the capacity to learn a structured skill systematically — to read a manual, follow a protocol, internalise an abstract procedure — was never constructed.

Specialisation produces the division of labour. The division of labour produces productivity. Productivity is wealth. This is the chain the introduction (Section 1) traced from Smith forward — Smith observed its terminal node and theorised wealth from the educated labour he took as given — and the chain Table 6 measures statistically: education → cognitive reorganisation →

specialisation → division of labour → productivity → wealth. Income is the last link. The residualisation finding — that income, stripped of education's contribution, predicts nothing — is the arithmetic consequence of that ordering. The presumed causal relationship between income and life expectancy, treated since 1975 as if income purchased health, was never that. It was education producing both, with income receiving credit.

The empirical confirmation. A direct test confirms the mechanistic argument. On the 85 countries where both completion rates and harmonised test scores exist (Angrist et al. 2021), I run a horse race: which measure better predicts life expectancy, TFR, and child mortality? With country fixed effects at a 10-year forward lag, completion explains 26% of within-country TFR variation; test scores explain 0.5% ($p = 0.48$). For child mortality the ratio is similar: completion $R^2 = 0.48$, test scores $R^2 = 0.03$ ($p = 0.24$). Hanushek & Woessmann's quality measure predicts GDP growth — their dependent variable. It does not predict development outcomes.

Adding duration of schooling to the horse race confirms the mechanistic claim directly. On the overlap sample where completion, test scores, WCDE mean years of schooling (age 20–24, weighted by cohort attainment), and Barro-Lee mean years (age 15–24, independent dataset) all exist, both duration measures beat test scores at a 10-year forward lag: for life expectancy, WCDE mean years $R^2 = 0.173$ and Barro-Lee mean years $R^2 = 0.076$ versus test scores $R^2 = 0.000$ ($p = 0.82$); for TFR, 0.338 and 0.101 versus 0.005 ($p = 0.47$); for under-5 mortality, 0.449 and 0.305 versus 0.028. Duration of exposure — measured two different ways, on two different datasets — predicts development outcomes. Fidelity of instruction does not.[2]

The debate over learning outcomes versus years of schooling (World Bank 2018; UNESCO 2023) measures the wrong dimension of the right variable. Test quality matters for GDP growth, which is Hanushek & Woessmann's dependent variable. It does not matter for the fertility transition, child survival, life expectancy, or intergenerational transmission — the outcomes that constitute human development. The empirical result (completion $R^2 = 0.26$ for TFR vs. test scores $R^2 = 0.005$) is not a puzzle. It is the mechanism's prediction: exposure to schooling reorganises cognition and opens the generational gate;

[2] See `scripts/robustness/completion_vs_years_vs_tests.py`.

test performance determines how much income the reorganised individual earns. Both are real. Only one is the cause of development.

4.4 Empirical Strategy

Each country is compared only to its own past. The question: does parental education predict what happens to their children one generation later?

Primary specification — country fixed effects regression:

$$E_{it} = \alpha_i + \beta_1 E_{i,t-25} + \varepsilon_{it}$$

In plain language: a country's education level today is predicted by its own education level 25 years ago (the parents' generation). The α_i term gives each country its own baseline — so the test is not "which countries are more educated?" but "within the same country, does the previous generation's education predict the next generation's?" This absorbs everything permanent about a country: institutions, geography, culture, colonial history. What remains is change over time. The 25-year lag is not a researcher choice — it is the time-to-agency interval derived from the biological mechanism (Prediction 1).

Comparison specifications replace parental education with income, or include both:

$$E_{it} = \alpha_i + \beta_2 \ln Y_{it} + \varepsilon_{it}$$

$$E_{it} = \alpha_i + \beta_1 E_{i,t-25} + \beta_2 \ln Y_{it} + \varepsilon_{it}$$

For development outcomes (life expectancy, fertility, child mortality), I measure outcomes 25 years after the education measurement to eliminate reverse causality — education at time T predicting life expectancy at time T+25:

$$O_{i,t+25} = \alpha_i + \gamma_1 E_{it} + \gamma_2 O_{it} + \varepsilon_{it}$$

Including the initial outcome level (O_{it}) is deliberately conservative: it absorbs the country's prior trajectory, so the test asks whether education predicts improvement beyond whatever trend was already underway.[3]

Residualization strategy. If education causes income, part of income's apparent predictive power is really education's effect in disguise. To test whether income contributes anything on its own, the design strips out the portion of income that education produced and uses only the remainder. Mechanics and results are in Section 4.7 (Table 6).[4]

On year fixed effects. Adding year FE collapses the parental coefficient to β=0.083 (p=0.07) on the full panel — a known pathology of staggered adoption. The Callaway–Sant'Anna estimator, designed for this exact problem, produces a clean positive effect (Appendix A, with a Goodman-Bacon decomposition documenting why the two-way FE specification breaks down).

4.5 Education vs. GDP as Predictors of Attainment

Where it matters most — countries still building their education systems — education predicts the next generation's attainment 3× better than income does.

Among countries where the expansion is still active — parental completion below 30%, where most of the remaining 20% of humanity sits — education R^2=0.70 vs GDP R^2=0.21. Education leads by 3.3×. GDP's R^2 barely changes across cutoffs (0.21–0.30); education's jumps from 0.53 to 0.70. GDP has no S-curve structure — its R^2 is flat across all cutoffs. Education leads GDP at every cutoff from 10% to 90%, with ratios ranging from 2.0× to 3.3×.[5]

Table 1. Country fixed effects regressions: child lower secondary comple-

[3]The main analysis uses aggregate (both-sexes) lower secondary completion. Female completion produces stronger R^2 on every outcome tested (Section 4.7); the aggregate is the more conservative specification, and the results reported here are a lower bound.

[4]Applied across three education levels (primary, lower secondary, upper secondary), four outcomes (life expectancy, TFR, child mortality, child education), and multiple sample definitions. See `scripts/residualization/education_vs_gdp.py`, `scripts/residualization/education_vs_tfr.py`, `scripts/residualization/child_education.py`. An entry-cohort design supplements the cutoff analysis: for each threshold (10% to 90%), identify the first year each country crosses it, then include all observations from that point forward while education remains below a ceiling (50%–90%). See `scripts/residualization/by_entry_ceiling.py`.

[5]See `scripts/residualization/by_gdp_cutoff.py`.

tion on parental education and log GDP per capita. Countries with parental completion below 30% (active expansion); 629 country-years, 105 countries.

Model	Parental edu β	Log GDP β	R^2 (within)
(1) FE: child ~ parent	**1.376*** **	—	**0.699**
(2) FE: child ~ log GDP	—	13.659***	0.214

*** *p<0.001. Standard errors account for non-independence within countries. GDP is not included alongside education in the same regression because income is on the causal path — controlling for it would mask education's effect (Section 4.1). The proper test strips education's contribution from GDP first (Table 6). Ceiling countries excluded. Education leads GDP at every cutoff from 10%–90% (`scripts/residualization/by_gdp_cutoff.py`).*

The decisive test is the shape across lags (Figure 4; Section 4.6).

Each generation gains more than it inherits (Prediction 6). The generational amplification coefficient β_g measures how much more educated the children's generation is relative to the parents'. If children merely inherited their parents' education level, β_g would be 1 or less. In fact it exceeds 1: at the <20% cutoff, β_g=1.032 (t=5.8); at <10%, β_g=1.019 (Table A1). The state adds reach on top of what households transmit.

The amplification is strongest where baselines are lowest: among countries below 20% completion, the next generation's gain is 2.86× the parents' level (education explains 70% of the variation); below 50%, 1.83× (explains 79%); below 90%, 1.24× (explains 77%).[6] The all-country panel (1900–2015) produces β_g=1.04 — stronger, not weaker, than the post-1975 panel. The long-run number pulls slightly higher because it reaches back into the low-baseline era, where state reach adds more on top of what PT alone transmits; the post-1975 panel is weighted toward countries already partway up the S-curve, where ceiling compression is starting to bite.

The S-curve within countries. Within individual countries, β_g varies systematically with baseline education level (Figure 2). The United States shows β_g=1.9 when its baseline was 13% (early 20th century), declining to β_g=0.08

[6] See `scripts/robustness/beta_by_ceiling_cutoff.py`.

when its baseline reached 92%. Korea shows β_g=6.5 at 1% baseline, declining through 3.6, 1.8, and 0.2 as its baseline rose to 59%. Taiwan follows a nearly identical trajectory (β_g=5.1 at 1.2%). The Philippines (Section 6) shows a flatter curve: β_g=4.4 at 1.5% declining more slowly to 0.4 at 49%. The S-curve structure is robust across all countries with sufficient historical depth. Bangladesh shows β_g≈2 throughout its observed trajectory (1–18% baseline), consistent with a country still in the steep part of the S-curve. India follows a similar pattern at β_g≈1.6 through 28% baseline.

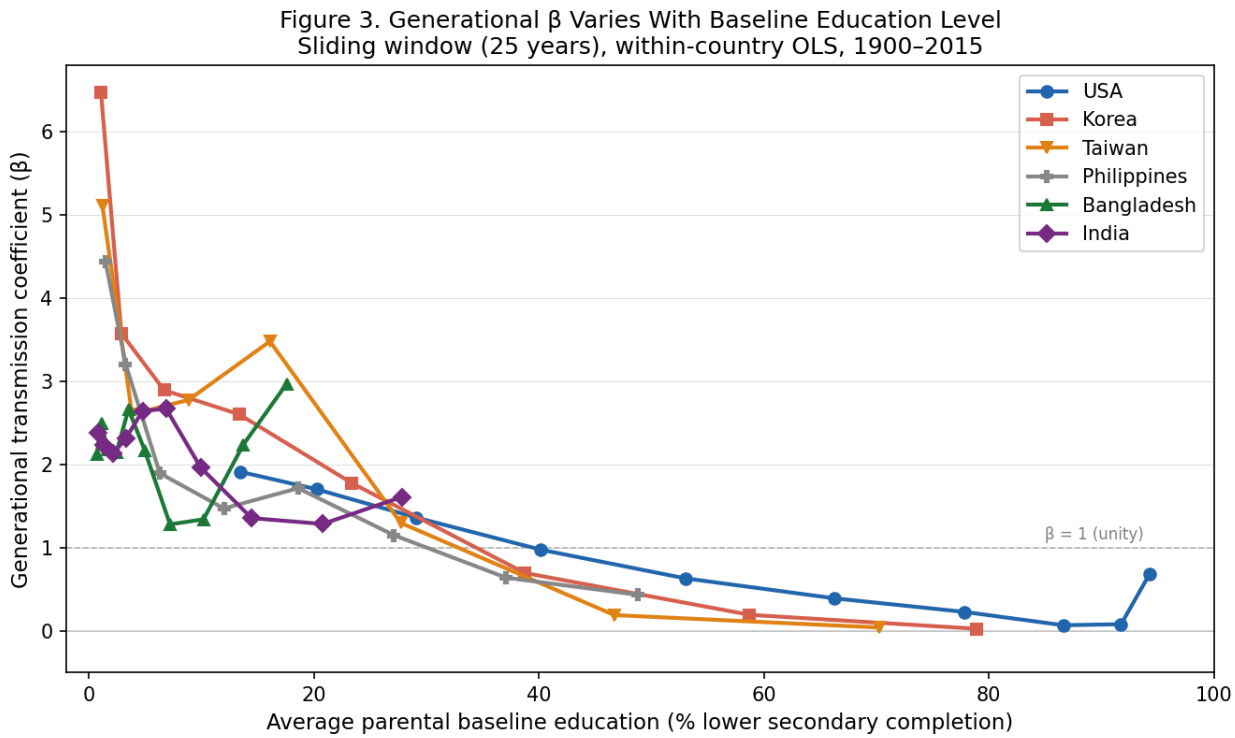

Figure 3: Generational β_g varies systematically with baseline education level. Each point represents a 25-year sliding window (6 child cohorts) within a single country, plotted against the average parental baseline for that window. $\beta_g > 1$ at low baselines reflects state reach; $\beta_g \to 0$ at high baselines reflects ceiling compression. *(Source: author's calculation, WCDE v3 long-run cohort reconstruction 1875–2015 — broader coverage than the 28-country regression panel in Table A5; Philippines historical data carries greater uncertainty for pre-1950 colonial-era observations. See* `scripts/figures/beta_vs_baseline.py`*)*

β_g approaching zero at high baselines is ceiling compression, not diminishing returns (distinct from Lutz & Kebede's cross-level finding that education's effect on life expectancy shows no diminishing returns from primary through tertiary). Non-depreciation holds universally — the Cambodia test (Section 5).

4.6 Education Predicts Development Outcomes 25 Years Forward

Knowing how educated a country's parents were predicts how long their children will live, how many children they will have, and how much they will earn — one generation later.

Table 3. Fixed effects regressions: development outcomes one generation forward on education, controlling for initial outcomes. Panel A reports education predicting development outcomes forward; Panel B reports the symmetric reverse regression (GDP predicting education forward) to demonstrate mutual endogeneity.

Panel A: Education(T) → development outcomes(T+25). n=828 (185 countries) for GDP row; n=1,295 (185 countries) for LE and TFR rows.

Outcome (one generation forward)	Education β	Initial outcome β	R^2 (within)
Log GDP per capita	+0.012***	+0.173***	0.354
Life expectancy (years)	+0.109***	+0.301***	0.382
Total fertility rate	−0.032***	+0.039*	0.362

The reverse regression tests whether GDP predicts future education with comparable strength — it does not once parental education is controlled.

Panel B: Log GDP(T) → education(T+25) — reverse regression

Outcome (one generation forward)	Log GDP β	Initial edu β	R^2 (within)
Lower secondary completion	+14.85***	—	0.272
Lower secondary completion	+3.780***	+0.485***	0.500

Country fixed effects. Panel A: n=828 (185 countries) for GDP row; n=1,295 (185 countries) for LE and TFR rows (GDP coverage is sparser). Panel B: n=828 for GDP-education matched obs.

Income alone explains 27% of child education variation across the full panel (Panel B row 1). But add the parents' education to the same model and income's apparent contribution is largely absorbed — the model's explanatory power jumps from 27% to 50%, almost entirely from education, not income.

Each 1-percentage-point gain in lower secondary completion predicts, one generation forward: 1.2% higher GDP per capita, 0.109 more years of life, and 0.032 fewer children per woman — all from within-country change over time, controlling for each country's starting point.

The education-vs-GDP predictive gap holds for child education at every cutoff tested.[7]

For life expectancy the gap is even wider: among countries below 30% completion, education explains 31% of the variation vs 2% for GDP — a 15× gap. Among countries below 10%, education explains 63% while GDP explains 1.6%. Even that small GDP signal vanishes once education's contribution is stripped out (Section 4.7).

Lag-decay shape (Prediction 2)

Education's signal survives four generations. Income's collapses within one.

Each lag corresponds to a generational relationship, and the relationship names what the number actually measures:

[7] See `scripts/residualization/by_gdp_cutoff.py`.

Lag (yr)	Gens	Relationship	R^2
0	—	same cohort (contemporary)	0.562
25	1	parent → child	0.364
50	2	grandparent → grandchild	0.171
75	3	great-grandparent → great-grandchild	0.085
100	4	great-great-grandparent → great-great-grandchild	0.052

R^2: within-country FE, life expectancy on lower secondary completion.

Each generation halves the remaining signal, but it never reaches zero: a great-great-grandparent's school completion still predicts how long their great-great-grandchild will live, four generations later — roughly a century of PT running through three intermediate households.

No other development input has comparable reach. Income's R^2 is already fading by lag 20–25, and it cannot be tested beyond that — not because the data is missing but by construction. Pre-literate subsistence societies had uniformly low income (~$400–600 per capita, constant 2017 USD), so there is no cross-country income variation to test before literate populations existed. The uniformity is not offered as evidence for the mechanism; it is a reminder of what the mechanism had to produce before income could become a variable at all. Institutions, quality scores, and provision fail the same generational-depth test: none survives at the depth education reaches. This is not a separate finding for each alternative — it is a single methodological principle derived from the mechanism itself. Backfilling pre-1960 income at $500 does not rescue income's predictive power.[8]

The grandparent channel

The lag-decay results show that education predicts across multiple generations. A sharper test: does the grandparent's education predict outcomes *independently*, after controlling for the parent's own education?

For fertility in the full panel, the answer is no — grandparent education adds nothing once the parent's is controlled (β_{gp}=+0.001, p=0.80). The parent absorbs the grandparent's entire contribution to fertility. But restrict to coun-

[8] See `scripts/robustness/le_r2_by_lag_backfilled.py`.

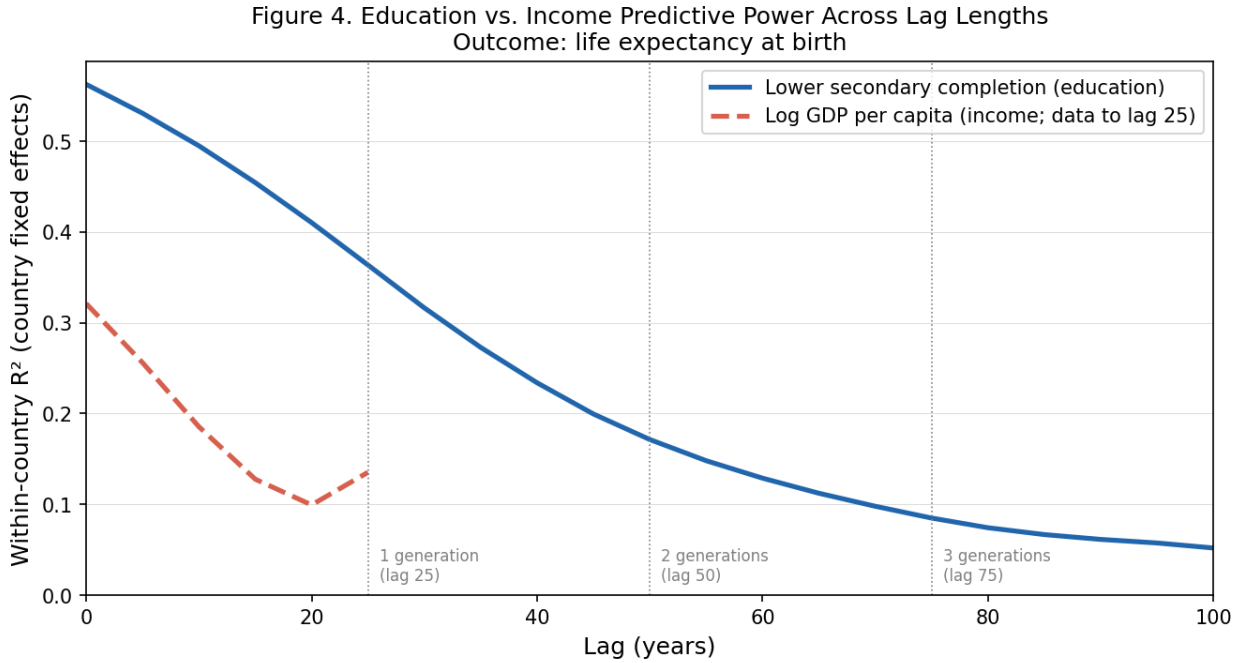

Figure 4: Education vs. income predictive power across lag lengths (within-country FE, outcome: life expectancy at birth). Within-country R^2 for lower secondary completion and log GDP per capita plotted against lag lengths from 0 to 100 years. The four generational horizons (lag 25 = 1 generation, 50 = 2, 75 = 3, 100 = 4) are marked. Income is shown through lag 25 only; beyond that, income was uniformly low across pre-literate subsistence societies, so the comparison is undefined. *(Source: author's calculation, World Bank WDI and WCDE v3. See* `scripts/figures/le_r2_by_lag.py`*)*

tries where education is still scarce (parent completion below 50%) and the grandparent effect emerges: β_{gp}=−0.059 (p=0.0003), nearly twice the parent's own coefficient (β_p=−0.033). R^2 jumps from 0.39 to 0.46. Where education is scarce, having an educated grandparent independently reduces fertility beyond what the parent's own education predicts. The effect is not gendered: grandfather and grandmother education produce comparable coefficients in the low-education subsample (β=−0.054 and −0.050 respectively), consistent with oblique transmission — in a village where education is rare, any educated elder is a visible exemplar.

For life expectancy, the grandparent effect is significant across all education levels (β_{gp}=+0.070, p<0.0001), adding 3.6 percentage points of R^2 beyond the parent alone. Roughly half the grandparent's total effect on LE runs through the parent; half is direct. For grandchild education, the effect is strongest: β_{gp}=+0.271 (p<0.0001), R^2 gain of 5.2 points.

The pattern is what the generational gate predicts (Section 4.3): at low baselines each educated person in the household adds independent weight; at high baselines the parent has internalised everything the grandparent could offer. The grandparent effect on fertility is real, but it operates precisely where it matters most — the steepest part of the transition.[9]

This generates a falsifiable prediction that distinguishes the kin/community-radius account from a pure household account: not only a living educated grandparent in the home, but any educated elder inside the child's daily kin/community radius — co-resident grandparent, village elder, aunt, older cousin — should raise grandchild outcomes beyond what the parent's education alone predicts, while equivalent elders present only on paper (deceased, non-resident, absent from daily life) should not. The channel runs through physical presence and daily interaction at Radii 1–2 (household and kin/identity; Section 2.3), not through inherited status. That the grandfather and grandmother coefficients come out nearly equal in the low-education subsample (β=−0.054 and −0.050) is consistent with the same reading: any educated elder works, not specifically the grandparent of the same sex. DHS co-residence data can test the presence-versus-absence prediction directly.

[9] See `scripts/robustness/grandparent_effect.py`.

4.7 GDP Has No Independent Effect (Prediction 4)

Take each country's income and remove the part that education produced. Use only what remains to predict development outcomes. Income predicts nothing.

GDP(T) correlates with future outcomes partly because education(T) drives GDP growth. The test: remove education's contribution to GDP and see whether the remainder predicts anything.

Table 6. Residualized GDP analysis: education vs. raw GDP vs. GDP after removing education's contribution. Entry-cohort design (entry ≥ 10%, ceiling ≤ 90%), country fixed effects, lower secondary completion, T=1960–1990, lag=25 years.

Outcome (T+25)	Edu R^2	Raw income R^2	Income R^2 (edu removed)	p-value
Life expectancy	0.472	0.165	**0.003**	0.56
Total fertility rate	0.478	0.175	**0.000**	0.98
Child education	0.524	0.303	**0.008**	0.31
Under-5 mortality	0.284	0.046	**0.023**	0.11

"Income R^2 (edu removed)" is the predictive power of income after stripping out education's contribution — the question is whether the leftover income predicts anything on its own. Education explains 42% of within-country income variation across all four outcomes; the remaining 58% predicts nothing. Standard errors adjusted for country clustering. n=702 (131 countries) for LE and TFR; n=856 (157 countries) for child education; n=734 (138 countries) for under-5 mortality. See `scripts/residualization/education_vs_gdp.py`, `scripts/residualization/education_vs_tfr.py`, `scripts/residualization/child_education.py`.

After removing education's contribution to GDP, income has no independent predictive power for any development outcome. Residualized GDP R^2 never exceeds 0.023; p-values all above 0.1. Education explains 42% of within-country GDP variation; the remaining 58% predicts nothing.

Child mortality: one bounded exception. Child mortality is the only out-

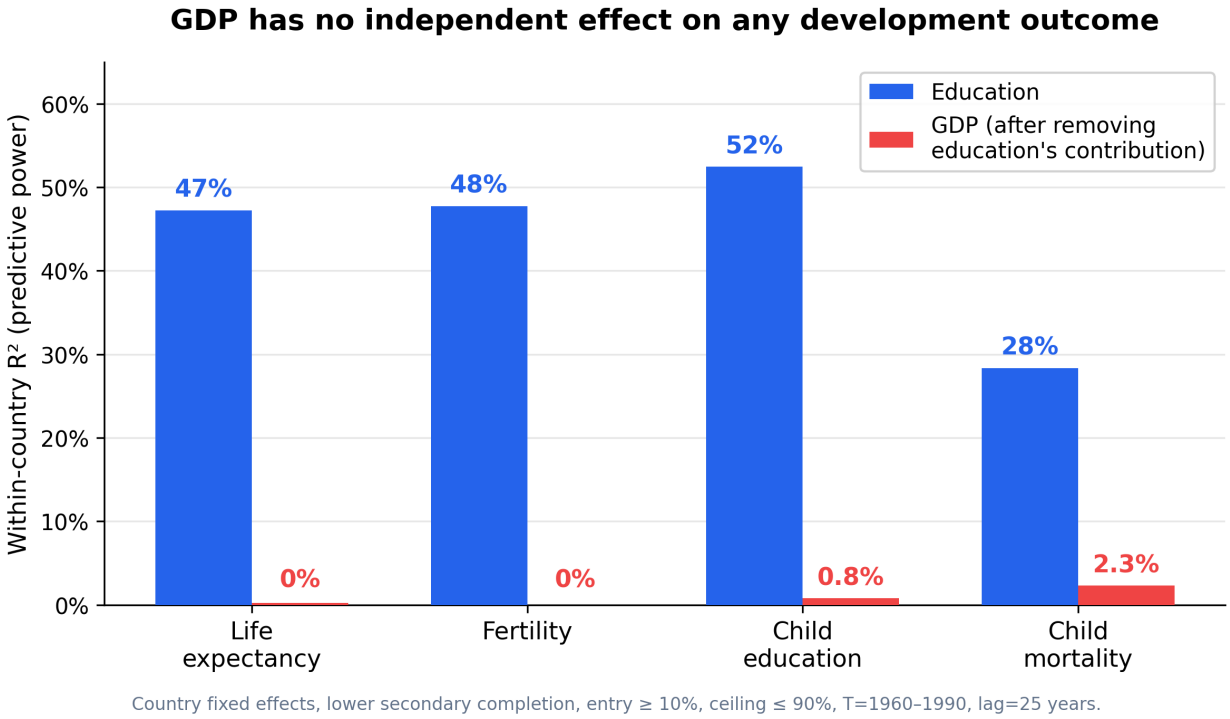

Figure 5: How much does education vs. income predict development one generation forward? Blue bars: education's predictive power. Red bars: income's predictive power *after stripping out what education produced*. Income predicts nothing on its own. Entry-cohort design (entry $\geq$ 10%, ceiling $\leq$ 90%), country fixed effects, lower secondary completion, T=1960–1990, lag=25 years. *(See* `scripts/figures/edu_vs_resid_gdp_bars.py`*)*

come where residualized income shows a marginal independent effect. Before 2000, leftover income explains 0.8% (not statistically significant); after 2000, 3.2% (p=0.04). The effect appears only at intermediate education levels and vanishes below 20% completion in any period. The paper's claim does not require every residual to be education: income, through external provision (GAVI founded 2000, vaccine delivery through global supply chains), can produce a small independent effect on child mortality in the post-2000 window where such provision scaled. Whether the surviving signal is genuinely income or is global education funded by donor surplus — vaccines developed, delivered, and paid for by already-developed populations — the channel is fragile by construction: it runs on donor altruism, not self-interest (Section 2.3, outer radius), and development assistance for health is already declining.[10]

Composition by education level

Primary education drives fertility decline; deeper education drives life expectancy gains.

The *composition* of the effect varies by level. Primary education is the strongest predictor of fertility decline (R^2=0.65 vs 0.48 for lower secondary, 0.32 for upper secondary): the first educated generation — the categorical jump from no schooling to literacy — is where fertility falls hardest, universally, across every country in the dataset regardless of culture, religion, or economic structure. Caldwell (1976) predicts fertility decline when children shift from economic assets to costs; if correct, the steepest decline should track economic modernisation, not primary education. It tracks primary education, in agrarian and industrial contexts alike. The mechanism is not economic calculation; it is cognitive capacity. Life expectancy gains accumulate with deeper education: upper secondary (R^2=0.48) predicts better than primary (R^2=0.41). This is the structure that Lutz & Kebede (2018) identified when redrawing the income–mortality curve with education on the x-axis: the fertility transition is triggered by the first level, while the education–health relationship tightens at higher levels.

Residualized GDP R^2 never exceeds 0.03 for life expectancy and is zero for

[10] See `scripts/robustness/u5mr_by_edu_level.py`.

fertility, at every level.

The education–life expectancy relationship shows no diminishing returns — linear from primary through tertiary (Lutz & Kebede 2018). Among 74 countries with lower-secondary completion above 85% in 2010, college completion carries a within-group correlation of r=0.45 with life expectancy — a 5.7-year gradient between lowest and highest college-completion quartiles among countries that have already solved the lower-secondary problem.[11]

Robustness and additional tests

Every robustness test confirms the residualization result (Table 6). Residualized GDP R^2 stays below 0.003 for life expectancy and fertility across all lag lengths tested; under-5 mortality reaches 0.023 (Table A3). The confidence intervals do not overlap: education R^2 [0.33, 0.59] vs residualized GDP R^2 [0.00, 0.04] (500-replication bootstrap). An alternative estimation method designed specifically to correct for bias in short panels (Anderson-Hsiao instrumental variables) produces results within 3–10% of the main estimates.[12] Results are unchanged when time-varying institutional quality scores (Polity5 polity2) are added (Appendix B.1). Female education produces stronger results than the aggregate on every outcome.

The parental income test asks a simple question: does family income predict child education independently, or only because richer parents tend to be more educated? Parental income alone predicts child education moderately (β=15.4, R^2=0.293). But add parental education to the same regression and income's coefficient collapses by 72% to β=4.3 (p=0.04), adding almost nothing (R^2=0.014) beyond education alone. Education's coefficient barely moves (β=0.475 from 0.553) while income's collapses. Income transmits through education, not vice versa.

The Asian Financial Crisis (1997–98) tests Prediction 3: GDP collapsed across five countries (Indonesia −14.5%, Thailand −8.8%, Malaysia −9.6%, Philippines −3.0%, Korea −6%). Lower secondary completion continued uninterrupted in all five; Indonesia gained 5.4pp while losing 14.5% of GDP. Income was destroyed; the educated population was not; development continued (see

[11] See `scripts/cases/college_le_gradient.py`.
[12] See `scripts/robustness/robustness_tests.py`.

`scripts/robustness/asian_financial_crisis.py`).

Once education is included, initial fertility barely predicts subsequent fertility one generation forward (Table 3, Panel A, row 3: initial outcome β=0.039). Demographic momentum does not drive the transition — educated women's decisions do.

4.8 Policy Over-Performers

Some countries are educating children far beyond what their parents' generation would predict. They are uniformly poor.

Table 8. Countries delivering education above generational baseline, 2015 FE residuals. The residual measures how far a country exceeded its own historical trend.

Country	FE residual above baseline	GDP per capita (2015)
Maldives	+34.9 pp	$9,645
Cape Verde	+26.3 pp	$3,415
Bhutan	+26.1 pp	$2,954
Tunisia	+25.5 pp	$4,015
Nepal	+17.8 pp	$876
Viet Nam	+16.0 pp	$2,578
Bangladesh	+15.8 pp	$1,224
India	+14.1 pp	$1,584

Nepal at $876 (2015), Bangladesh at $1,224, Vietnam at $2,578. Income is not the prerequisite for educational over-performance.

4.9 The Shock Test (Prediction 3)

Exogenous shocks that kill people depress life expectancy. They do not reverse the fertility transition. The pattern holds across every shock type in the dataset.

Table 10. Exogenous shocks and development outcomes.

Shock type	Country	LE effect	TFR effect	Edu effect
Civil war	Sri Lanka	Delayed 12 yr	On schedule	None
State collapse	Russia	−5.0 yr, recovered	Unaffected	None
Pandemic (HIV)	South Africa	−9.0 yr, recovering	Continued falling	None
Income destruction	AFC (5 countries)	None	None	Continued rising
Population destruction	Cambodia	Stalled	Delayed	**Reset**

LE effect: deviation from pre-shock trajectory. TFR effect: deviation from education-predicted path. Sources: World Bank WDI, WCDE v3. See `scripts/verify_russia_south_africa.py` *for Russia and South Africa;* `scripts/robustness/asian_financial_crisis.py` *for the AFC; Sri Lanka and Cambodia in Sections 14 and 5.5.*

TFR is a household decision running on PT — internal to the household, unreachable by war, virus, or collapsing state. LE depends partly on whether external conditions allow survival. When the two diverge — TFR continuing its education-predicted decline while LE collapses — the divergence identifies the shock as external rather than educational.

Russia (99% lower secondary completion, 1990). The Soviet dissolution crashed life expectancy from 69.5 to 64.5 (1988–1994) — 5 years lost in six. TFR, already at 1.89, continued below replacement throughout — 1.20 by 2000, never reversing. The state dissolved; the educated population did not. LE returned to its pre-collapse level by 2009 (68.7) and reached 73.1 by 2019 — surpassing the Soviet-era peak by 3.6 years. Education predicts the recovery trajectory, not immunity to state collapse.[13]

South Africa and the HIV epidemic. South Africa's 65% lower secondary completion in 1990 is an aggregate masking apartheid stratification: near-universal white completion and much lower Black completion produced a bimodal distribution invisible in national data. The population bearing the brunt of HIV had far lower education than the headline figure suggests. HIV crashed

[13] See `scripts/verify_russia_south_africa.py`.

LE from 62.9 to 53.9 (1990–2005) — the largest peacetime LE reversal in modern history. TFR fell from 3.72 to 2.41 over the same period, uninterrupted.

The Southern African epidemic was driven primarily by factors education could not have prevented: low traditional circumcision prevalence (8–11% in the HIV belt vs 85–99% in Muslim West Africa), which three randomised trials showed provides a 60% biological reduction in male acquisition risk (Auvert et al. 2005); and circular mining migration that created connected sexual networks across the subcontinent. These are genuinely exogenous to education.

But education was in the story from the beginning — on both sides. Early in the epidemic, more educated Africans were *more* likely to be HIV-positive: greater mobility, more partners, more agency exercised without information about a novel pathogen (Fortson 2008). Once information existed, the gradient reversed: educated populations changed behaviour first. In Botswana, a natural experiment exploiting a 1996 school reform showed that each additional year of secondary schooling *caused* an 8.1-percentage-point reduction in cumulative HIV risk (De Neve et al. 2015). In rural Uganda, educated individuals responded more to prevention campaigns, and the education–HIV gradient reversed from positive to negative between 1990 and 2000 (De Walque 2007). Baker et al. (2017) formalise this as the Population Education Transition curve: when a novel health risk enters a population, the educated are early adopters (positive gradient); once information exists, the educated change behaviour first (gradient reverses). The pattern replicates across cigarettes, processed food, and HIV — education is the mechanism in both directions.

The fertility transition confirms the household channel. TFR runs on PT — internal to the household, unreachable by a pathogen that operates through sexual networks. LE depends on whether external conditions allow survival. Post-ARV, LE recovered to 66.1 by 2019. The TFR–LE divergence during the HIV crisis is itself diagnostic: the household mechanism continued; the external shock could not reach it.[14]

Sri Lanka (Section 14) and the Asian Financial Crisis (Section 4.7) confirm the pattern from different angles: civil war delayed LE without breaking the fertility trajectory; income destruction left both outcomes untouched.

[14] See `scripts/verify_russia_south_africa.py`.

Cambodia (Section 5.5) is the exception that proves the rule: the only shock that breaks the household mechanism is the one that destroys the household.

4.10 The Famine Test

The shock test asks whether education survives crisis. The famine test asks the converse: does education *prevent* the most extreme form of crisis?

Every major famine since 1950 occurred in a low-education setting. Of 21 famines in the dataset, 19 occurred where lower secondary completion was below 50%. The median education level at the time of famine was 19.6%; the mean was 25.4%. The two exceptions — North Korea (1996, 100% completion) and Yemen (2018, 68%) — both required external force physically preventing food access: totalitarian state control of distribution in the first case, naval blockade in the second. Where the educated population retained agency, famine did not occur.

The near-miss cases make the pattern sharper. Seventeen countries faced shocks comparable to or worse than those that produced famines elsewhere — Cuba's 35% GDP collapse when Soviet food imports were cut by 80% (1993), post-WWII Japan and Germany, Armenia under simultaneous blockade and war (1993), Ukraine under Russian invasion and grain blockade (2022) — and none experienced famine. Their median education was 71.6%, against 19.6% for famine countries. The difference is significant at $p < 0.00001$ (Mann–Whitney U).

Educated populations do not starve. They reorganise supply chains, ration collectively, substitute crops, demand state response, and migrate strategically. Uneducated populations, facing the same shock, lack the household-level capacity to adapt and the political capacity to compel response. The mechanism is PT operating through the household: the same cognitive and behavioural repertoire that drives the fertility transition also drives famine prevention. The test assumes the energy substrate (Section 2.1): every shock in the dataset was a distribution failure, not absolute absence of food in the region. Educated populations respond by solving the distribution problem; uneducated populations cannot.

Bihar and Kerala, 1966. The within-country case eliminates every con-

founder except education. The 1965–66 national drought crashed India's grain production by 19%. Bihar experienced famine: 70,000–130,000 excess deaths (Dyson & Maharatna 1992). Kerala — India's most food-deficit state, 40% import-dependent, hit so severely that food riots erupted and the UK Parliament identified it as one of the "most severely hit areas" — had no famine. Same Constitution, same free press, same central government, same democratic institutions. Bihar's literacy was 22% (female: 9%); Kerala's was 55% (female: 39%). (The 1966 figures are literacy rates — the only education statistic available at state level for that year — not the lower secondary completion measure used elsewhere in the paper; the gap runs in the same direction under any available measure.)

Sen's claim that "no famine has ever taken place in a functioning democracy" (Sen 1999) cannot explain the Bihar–Kerala divergence. Democracy was constant across both states. Education was not. Myhrvold-Hanssen (2003) challenges Sen directly on this point, arguing that Bihar 1966–67 meets any reasonable definition of famine and that the Indian media, "although free and independent, did not provide reliable information." Drèze himself (quoted in Brass 1986) described the evidence that Bihar was a success story as "precious little."

Kerala's own history confirms the mechanism. In 1943, Travancore (pre-independence Kerala) experienced famine under the same structural vulnerability — extreme food-import dependence — killing approximately 90,000 people. The difference between 1943 and 1966 was not democracy (Travancore had an elected legislature from 1932) but education: literacy had risen from approximately 30% to 55%. The educated population built the institutions — universal public distribution, food committees, political mobilisation — that prevented the same structural vulnerability from producing the same outcome.[15]

5. The Cases

Every country that developed did so one generation after sustained education investment — under every political system, at every income level, on every

[15] See `scripts/famine_education_test.py`.

continent.

Table 12. Year of development (TFR < 3.65 AND life expectancy > 69.8) and relationship to prior education investment. Thresholds are 1960 United States values (World Bank WDI).

Country	Dev.	TFR crossed	LE crossed	Onset	Rate	Lag	Gen.
Taiwan	**~1970**	~1970	~1970	1950s	2.15	~20 yr	**1**
S. Korea	**1987**	1975	1987	1953–65	2.13	~25 yr	**1**
Cuba	**1974**	1972	1974	1961 (40%)	2.27	~13 yr	**1**
Bangladesh	**2014**	1995	2014	1990s	1.30	~24 yr	**1**
Sri Lanka	**1993**	1981	1993	1940s–50s	1.20	~42 yr	**2**
China	**1994**	1975	1994	1950s+CR	1.50	~42 yr	**2**
Kerala†	**~1982**	~1973	~1981	Early 20thC	—	~65 yr	**3**
India	**2017**	1996	2017	1950s	0.87	~67 yr	**3**
Uganda	**—**	TFR 4.39	LE 67.7	None	—	—	—

Developed = year both thresholds crossed. Expansion onset = year or level at which sustained educational expansion began. Rate = average annual change in lower secondary completion from expansion onset to development crossing, in percentage points per year (see `scripts/expansion_rate_predicts_crossing.py`*). Lag = years from onset to development crossing. Generations = number of ~25-year cycles. Cuba's high starting base (40%) and the 1961 reconstruction of losses from the post-revolution exodus shortened the lag below the typical ~25-year generation. † Kerala figures estimated from India Sample Registration System and census records; all other figures from World Bank WDI direct measurement.*

The earliest modern example is Japan (1872): the Meiji compulsory education ordinance, implemented through repurposed temples with community teachers, no substantial budget. The sequence was mandate before resources — the state declared compulsory education and built the infrastructure afterward. Japan industrialised within a generation of these reforms despite negligible domestic fossil fuels, demonstrating that the mechanism does not require natural resource endowments.

Korea is the calibration case: 2.13 pp/yr, the fastest sustained expansion in the WCDE dataset. All other rows in Table 12 are independent tests. The expansion rate determines generations to crossing (Section 2.2): benchmark

pace compresses it to one generation; India's 0.87 pp/yr spreads it across three.

Three parameters predict crossing time: starting base, expansion rate relative to the Korea benchmark, and structural disruption. Disruption delays LE crossing — Sri Lanka's 12-year TFR-to-LE gap (1981→1993) maps onto the civil war; China's 19-year gap (1975→1994) reflects Deng's dismantling of the barefoot doctor network. Korea, Taiwan, Cuba, and Bangladesh had no comparable disruption. The predicted depths match: Korea-pace cases in 20–34 years; half-Korea-pace in 42–45 years; third-Korea-pace in 60–70 years. The crossing dates are the test, not the inputs.

5.1 Taiwan and Korea

Taiwan and Korea crossed earliest — both at benchmark pace, both state-driven from colonial bases built by Japan (25% and 18% completion by 1950). Korea expanded at 2.13 pp/yr, the fastest sustained rate in the WCDE dataset; Taiwan followed a nearly identical trajectory. Both had functioning markets — what Sen calls "growth-mediated security" — but the growth was mediated by education, not the reverse. State-driven singular expansion compressed development to 20–34 years against the 40–70 years seen in competing-priority cases.

American colonial education in the Philippines was similarly deep — 22% lower secondary completion by 1950, comparable to Korea. In 1960, the Philippines was one of the most prosperous countries in developing Asia: GDP per capita of $1,124 against Korea's $1,038 (constant 2017 USD), ahead of Thailand ($592), Indonesia ($598), and far ahead of India ($313) and China ($241). The Philippines had both the educational base and the income advantage. But no post-independence government sustained the educational investment. Korea made education the singular priority and expanded at 2.13 pp/yr; the Philippines drifted at roughly half that pace. The result is visible in Figure 3: comparable colonial base, comparable income, divergent post-colonial trajectory. Education policy is a choice, not an inheritance. Income does not make the choice for you.

5.2 Kerala

Kerala crossed at ~1982 — the paradigm case of competing-priority expansion. Education had been building since the early twentieth century through social reform movements — gradual accumulation through social reform and state investment — competing-priority, not singular-priority at Korea pace. TFR crossed by ~1974; LE followed ~8 years later. Kerala had extensive state provision — public health, food distribution, land reform — but fertility declined because educated women made the decision, not because provision was available (Section 2.6). The Emergency sterilisations (1975–77) confirm the polarity: forced on the least-educated northern states, they brought down the government (Gwatkin 1979; Vicziany 1982); Kerala needed no coercion. State coercion either arrives after education has done the work, or fails where education has not.

The sub-national test has been run systematically. Drèze & Murthi (2001), using district-level panel data for 1981 and 1991, found that women's education was the single most important predictor of fertility differences across Indian districts, while urbanisation, poverty reduction, and male literacy showed no significant association. The cross-country result replicates within a single country at district level.

5.3 Four Further Cases

Case	Crossed	Mechanism tested
Sri Lanka	1993	TFR crossed 1981; LE climbed to 69.0 by 1988, fell to 67.3 during the civil war, recovered to cross 69.8 in 1993. War disrupted the economy and life expectancy without breaking PT.
Myanmar	—	73 years of military rule, GDP $1,025 (2015), education at 0.6 pp/yr. TFR fell from 5.9 to 2.3; LE rose from 44.1 to 65.3 — consistent with 43.5% completion. The regime suppressed speed; it could not reverse what PT had transmitted.

Case	Crossed	Mechanism tested
Cuba	1974	40.3% completion in 1960; post-revolution exodus removed the professional class. The 1961 campaign deployed 268,000 volunteers (Prieto 1981), replacing an elite that left with a mass baseline that stayed. Cuba and Taiwan — Soviet-aligned and US-allied — crossed within four years.
Bangladesh	2014	11.4% completion in 1960, GDP $1,159 at crossing. Sustained commitment to girls' education from the 1990s. TFR decline tracks girls' secondary expansion, not contraceptive distribution (Mamun & Bongaarts 2022).

5.4 China

China crossed in 1994. Three corrections to the standard narrative.

First, what is called an educational catastrophe was the largest educational expansion in Chinese history. The standard framing conflates the disruption of university education — a small percentage of the population — with the overall trajectory. For rural China — 80% of the population (National Bureau of Statistics 1982) — the Cultural Revolution era produced the largest lower secondary gains in the WCDE 1870–2015 record: +10.6pp for the 1975 cohort, +15.0pp for the 1980 cohort. Community schools (民办学校) brought secondary education to villages that had none (Pepper 1996; Unger 1982; Gao 2008).

Second: China is the most direct test of Drèze and Sen's support-led-security thesis, and the thesis fails. Drèze & Sen (1989, Table 10.6) show China's under-5 mortality at 32% of what GNP predicted — the largest deviation in their sample — and attribute it to "support-led security": barefoot doctors (赤脚医生), direct provision. But Sen regressed on income and found a residual. That residual is education.

When life expectancy and under-5 mortality are matched on mean years of

schooling (age 20–24, ± 0.5 years of schooling, any calendar year) rather than GNP, China's advantage reverses. China's LE was *below* education-matched peers from 1965 to 1991 — 6.3 years below at 1965, still 2.4 below at 1980, converging only in 1991 — and its U5MR was *above* education-matched peers until 2000.[16] The barefoot doctors were not producing exceptional outcomes; China was underperforming what its education level predicted.

The barefoot doctors and the community school teachers came from the same CR-era rural mobilisation (Gao 1999). The doctors were local, culturally embedded, drawn from the villages they served — optimally delivered provision. When Deng dismantled the system after 1980, they left, because rational people freed from revolutionary ideology redirected their labour toward self-interest. The LE convergence rate was statistically unchanged: $+0.28$ years/year before 1981, $+0.29$ after ($\beta_3 = +0.007$, $p = 0.78$). LE continued to rise — from 64 to 69.8 by 1994 — because the educational baseline built into rural households did not leave with the doctors. Drèze & Sen (1989, pp. 215–221) document the resulting health crises and rising rural medical costs (Gao 2008; Liu et al. 2003), but these crises left no detectable mark on the trajectory.

China crossed to *above* education-predicted LE in 1992 and to *below* education-predicted U5MR around 2000 — two decades after the barefoot doctors were gone. Mean years of schooling rose from 5.9 (1965) to 8.0 (1980) to 9.6 (2000), tracking both outcomes across both eras.

Third, the TFR threshold was crossed in 1975 — five years before the one-child policy. China's fertility trajectory mirrors South Korea and Thailand's, both of which achieved equivalent TFRs without compulsory policy (Cai 2010; Miller et al. 2018).

5.5 Cambodia: The PT Shadow

Cambodia is the direct test of Prediction 3: what happens when the educated population is destroyed. The country experienced severe disruption under the Khmer Rouge (1975–1979), during which the education system collapsed.

[16] See `scripts/cases/china_mean_years.py`. Mean years of schooling computed from WCDE v3 proportions (age 20–24, both sexes); LE and U5MR from World Bank WDI. Peer pool is all country-years within ± 0.5 mean years of schooling of China's value at each year.

The data shows two distinct stalls. The first is the exogenous shock itself. The second — a generation later — is PT's prediction: the household baseline is the binding constraint (Section 2.2, PT mechanism).

1975	Khmer Rouge takeover; completion 10.1%
1979	Khmer Rouge falls; educated population destroyed
1985	Completion 9.5% — **first stall** (system collapse)
1993	Paris Accords; UN transitional authority ends
1995	International reconstruction; completion jumps to 35.1%
2010	Plateau at 31–36% — **second stall** (parental baseline)
2011	Recovery as post-disruption cohort's children enrol

First stall (1975–1985). Completion was 10.1% in 1975, fell to 9.4% by 1980, and was still 9.5% in 1985 — a decade of zero progress.

Recovery and second stall (1990–2010). After the Paris Accords and UN transitional authority (1991–1993), schools were rebuilt with sustained international investment. Completion jumped to 35.1% by 1995 on the back of that reconstruction. Then it stalled again. From 1995 to 2010, completion plateaued at 31–36%, unmoved despite continued external funding. The buildings were there; the teachers were there; the money was there. Progress did not come. The 35% plateau is what competing-priority expansion looks like when it interacts with a parental education shadow: international reconstruction dispersed resources across health, infrastructure, governance, and education simultaneously, while the parental cohort remained frozen at ~10%. Korea proves singular priority can push far beyond any parental baseline — but Cambodia after 1991 was not Korea. The plateau reflects dispersed investment meeting a damaged generational base.

The first stall is the exogenous shock — the regime destroyed the education system. PT predicts the second. It is the generational shadow: the children reaching secondary school in 1996–2010 were born approximately 1982–1996 — their parents were the cohort whose own education was frozen at ~10% during 1975–1985 — the pre-disruption level, preserved in households even as the school system collapsed. Countries that started at Cambodia's level in 1960 reached a median of 21% by 1985. The children of the frozen cohort inherited a 10% parental baseline instead of a ~21% one; the plateau reflects that missing growth. Recovery from 2011 onward corresponds to the

post-disruption cohort's children finally dominating the school-age population. The regime fell in 1979, but education remained frozen through 1985 (9.5%). Twenty-five years from the end of the educational disruption (1985) lands at 2010 — precisely when recovery began.

Countries that started at Cambodia's level in 1960 reached a median of 46% by 2015. Cambodia reached 36%. The buildings came back in 1991. Progress came back in 2011. That twenty-year gap is the PT shadow.

The shadow is not over. The grandparent channel (Section 3) predicts a second, deeper lag. Children reaching adulthood in 2020–2025 have parents from the post-reconstruction cohort (35–36% completion) but grandparents from the frozen cohort (9–10%). At low baselines, grandparent education predicts child outcomes independently of parent education, with a coefficient nearly twice the parent's ($\beta_{gp} = -0.059$ vs $\beta_p = -0.033$ for fertility). The Khmer Rouge fell in 1979; the parental shadow lifted in 2011; the grandparent shadow persists through at least 2035, when the first post-disruption cohort's children become grandparents themselves.

6. The Decision

Every country that developed made the same decision: sustained investment in education. The only variation is speed.

Spain is the longest counter-case to the wealth-first reading: one of the wealthiest empires in history, it had 0.3% lower secondary completion for the 1900 birth cohort and did not reach universal completion until approximately 2000. Wealth was present for 450 years. Mass education took that long to arrive. Korea compressed the same transition to one generation. The mechanism is the same; what varies is how fast the state acts.

The state is necessary for scale but not for efficacy. PT operates at the household level regardless of state form — autocracy, democracy, colonial administration, revolutionary government. What the state provides is reach: the extension of education to children whose parents have none. The state is the delivery mechanism, not the cause. The era of no priority — the pre-industrial baseline where hierarchy (de Waal 2013) suppressed mass education — is over. The global education signal is now visible everywhere: every

government on earth claims to prioritise education (UN 2015, SDG 4; UNESCO GEM 2024). The question is no longer whether to educate but at what speed. Competing priority is the norm. The decision this paper calls for is singular priority.

The question for the remaining 20% is whether evidence can now substitute for the contingent motivations that drove every previous breakthrough. Knox, Meiji, and Korea bypassed the cost-benefit frame by never entering it; the commitment preceded any calculation. Evidence-based commitment would bypass the frame from the other direction: by showing that the calculation has already been done across 185 countries and the answer is always the same — education first, everything else follows.

The function of the evidence is not to win a cost-benefit comparison against health or nutrition, which education will always lose at decision-relevant timescales, but to remove the comparison altogether: to establish that education is not one investment among many but the precondition for all others.

This framework makes a falsifiable prediction.

Uganda reached 48.8% lower secondary completion in 2025, expanding at approximately 6 percentage points per five-year period. Its TFR is 4.39 and life expectancy 67.7 — below both development thresholds. If the framework is correct, Uganda will cross both thresholds (TFR < 3.65 and LE > 69.8) by approximately 2050, conditional on maintaining current expansion rates. The 25-year forward lag from 2025 education levels predicts outcomes in 2050; the grandparent channel (Section 3) predicts the crossing will be modestly delayed by Uganda's low grandparent baseline (9.1% in 1975 for the 2025 cohort). If Uganda crosses on schedule, the framework is confirmed in a case it did not train on. If it does not, and expansion was sustained, the framework is wrong.

One scope note, since the question is often raised: the material question that haunted every earlier call for universal development — whether the earth's energy base could sustain decent lives for everyone — is being resolved in parallel with, but independently of, the development decision. The solar–wind–battery transition is changing the cost and scale of energy capture on a trajectory that relieves the forward-looking resource constraint (Smil 2017). This

paper takes no position on its speed. The point is structural: the substrate is becoming tractable without further political choice in each country where it is needed. The variable that still requires a choice, in every country where development has not arrived, is education (Section 2.1).

No leader has yet made the decision on the basis of evidence alone. This paper provides the evidence. The decision remains.

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Wolfgang Lutz's datasets made this analysis possible and his work on education and human capital set the terms on which it stands. He will recognise the question this paper is answering as his teacher's.

Richard Wrangham established what the human species was before education reached it — what cooking made, what violence shaped, what self-domestication permitted.

Sarah Blaffer Hrdy established that the household which transmits is never just two parents. The cooperative breeding that made the eighteen-year dependency period viable is the precondition this paper inherits.

Melvin Konner's *The Evolution of Childhood* is the biological foundation this paper builds on. The eighteen years that separate the human juvenile period from every other species is the fact from which everything else follows. *The Tangled Wing* has been a longer, quieter presence.

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Frans de Waal spent his life showing what primates reveal about what is possible — the bonobo's imperfect, conflicted cooperation alongside the chimpanzee's lethal hierarchy. Without the first the household does not hold; without understanding the second we cannot see what education overcomes. He is missed.

Hunger and Public Action was one of the most important books this author read. Working with Jean Drèze to make it freely available was a privilege.

Drèze and Sen saw the right anomalies and asked the right questions. This paper is one answer.

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Development threshold data computed from World Bank WDI. USA 1960 benchmarks: TFR = 3.65, life expectancy = 69.8. Full analysis code, data, and verification outputs available at github.com/rkpagadala/education-of-nations.

A. Technical Robustness

Table A1. Two-way fixed effects (country + year): child lower secondary completion on parental education. R^2 is the partial within-R^2 (incremental variance explained beyond country + year FEs). See `scripts/robustness/twfe_child_edu.py`.

Model	Parental edu β	R^2 (within)	n	Countries
(1) FE + year: parent (< 30%)	**0.739*** **	**0.154**	783	137
(2) FE + year: parent (< 20%)	1.032***	0.145	600	118
(3) FE + year: parent (< 10%)	1.019***	0.067	358	85
(4) FE + year: parent (all)	0.083	0.009	1,665	185

*** *p<0.001. Standard errors account for non-independence within countries. Model (4) does not reach conventional significance (p=0.07) — see explanation below.*

Why Model 4 shows β=0.083 and why it does not matter. When each country is compared only to its own past (one-way fixed effects), a 10-point increase in parental education predicts a 4.8-point increase in child education one generation later (β=0.483). Adding year fixed effects — which also control for which *decade* an observation falls in — collapses this to β=0.083. The question is whether this collapse reflects a real weakness in the evidence or a flaw in the statistical method.

Recent econometrics has shown it is the method. Goodman-Bacon (2021) proved that when you add year controls to a panel where countries adopted education at different times, the result is a weighted average of every possible before-and-after comparison between pairs of country groups. Some of these comparisons are clean: countries that expanded education compared against countries that had not yet started. Others are contaminated: countries just beginning their expansion compared against countries already in the middle of theirs — as if measuring whether a runner accelerated by comparing her

to someone already sprinting.[17]

Models 1–3 partly address this by restricting to countries still expanding education, where the effect is strong. The Callaway–Sant'Anna (2021) estimator addresses it directly: it compares each country only against countries that have not yet started expanding, eliminating the contaminated pairings altogether. The result: crossing the 10% education threshold produces a gain of 7.9 percentage points in child education one generation later (95% CI: 4.4–11.0).

The event study (Figure 6) makes the pattern visible. Before a country crosses the 10% threshold, there is no effect — countries that will expand and countries that will not are on identical trajectories. After crossing, the effect grows steadily: +1.3 percentage points immediately, +6.6 at ten years, +14.9 at twenty-five years, +21.4 at thirty-five years. Parents who cross the education threshold raise more educated children, who raise still more educated children. The effect compounds.

Table A3. Robustness to alternative lag lengths. Education R^2 vs. residualized GDP R^2 across lags 15–30 years. Entry-cohort design (entry $\geq$ 10%, ceiling $\leq$ 90%), country fixed effects, lower secondary completion. The T+25 specification is the theoretically motivated anchor, not a cherry-picked optimum; the result holds at every lag tested. See `scripts/robustness/lag_sensitivity.py`.

	Life exp.		TFR		U-5 mort.		Child edu	
Lag	Edu	Resid	Edu	Resid	Edu	Resid	Edu	Resid
15 yr	0.463	0.000	0.572	0.000	0.420	0.007	0.725	0.011
20 yr	0.460	0.002	0.508	0.000	0.336	0.018	0.617	0.009
25 yr	0.472	0.003	0.478	0.000	0.284	0.023	0.524	0.008
30 yr	0.451	0.000	0.458	0.001	0.234	0.007	0.461	0.004

"Edu" = how much of each country's outcome change is predicted by education. "Resid" = how much is predicted by income after stripping out educa-

[17] A formal decomposition of the education panel confirms this: the clean comparisons show large effects; the contaminated ones show near-zero effects; and the contaminated comparisons dominate the average. See `scripts/robustness/goodman_bacon_decomposition.py`.

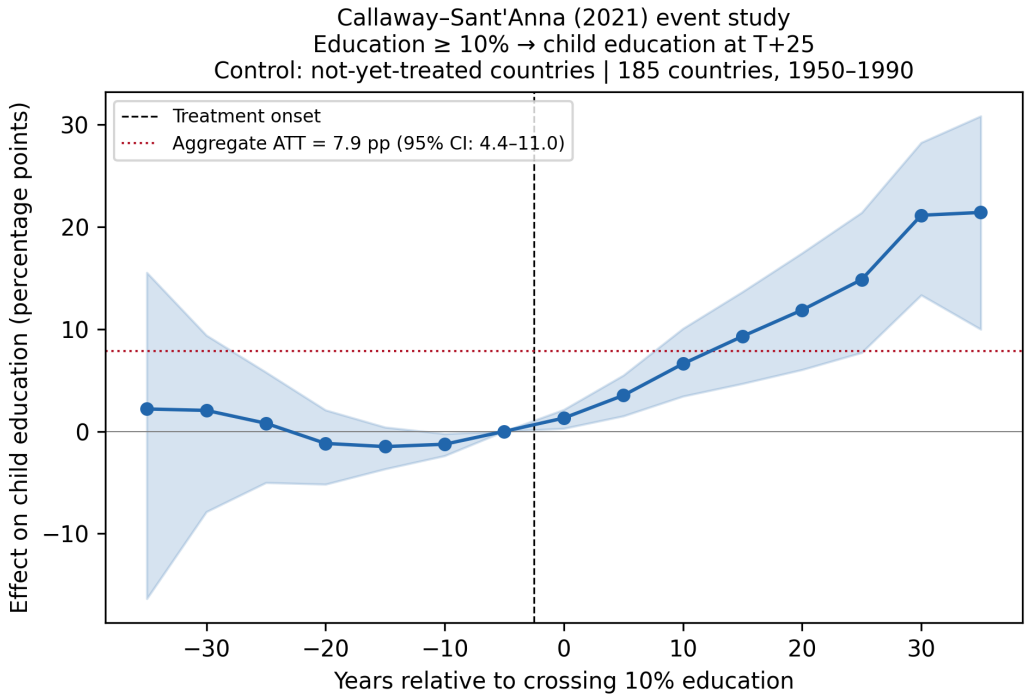

Figure 6: Callaway–Sant'Anna (2021) event study. Treatment: country crosses 10% lower secondary completion. Outcome: child education at T+25. Control group: countries not yet treated at each comparison point. Shaded band: 95% confidence interval (500-replication clustered bootstrap). Pre-treatment effects near zero confirm parallel trends. Post-treatment effects grow monotonically from +1.3 to +21.4 percentage points over 35 years — consistent with compounding intergenerational transmission. *(See `scripts/robustness/callaway_santanna.py`.)*

tion's contribution. At every lag tested and every outcome, education explains 23% or more of the variation; leftover income never exceeds 2.3%. n=702 (131 countries) for LE and TFR; n=734 (138 countries) for U-5 mortality; n=856 (157 countries) for child education.

Table A5. Long-run panel countries (1900–2015, 28 countries). Selected on the criterion of self-determined education policy and data quality; colonial-era data for colonised countries reflects colonial investment rather than domestic policy decisions and is excluded.[18]

Table A6. Development threshold robustness: crossing dates under three specifications (World Bank WDI data; Taiwan excluded — not in WDI). The loose spec uses generous thresholds; the main spec uses the 1960 USA values; the strict spec tightens to TFR < 2.5 and LE > 72.6 (USA 1975).

Case	Loose	Main	Strict	Shift
Cuba	1964	1974	1980	16 yrs
South Korea	1978	1987	1993	15 yrs
Sri Lanka	1975	1993	2005	30 yrs
China	1982	1994	2001	19 yrs
Bangladesh	2005	2014	2019	14 yrs

Note: Every country crosses under every specification, and the ordering never changes. The threshold choice shifts dates by 14–19 years for fast expanders (Korea 2.13 pp/yr, Cuba 2.27 pp/yr, China 1.50 pp/yr) and 14–30 years overall, but no country that crosses the loose spec fails to cross the strict one. Sri Lanka's 30-year shift reflects the civil war holding life expectancy near the threshold while the educational base accumulated beneath it; the binding constraint was disruption, not the threshold definition. Taiwan (1970 loose, 1980 strict; WCDE data) is consistent with Cuba under all specifications but cannot be computed from WDI data. See `scripts/robustness/threshold_robustness.py`.

[18]The 28 countries: *East Asia* — Japan, Taiwan, South Korea. *Western Europe* — United Kingdom, France, Germany, Sweden, Norway, Denmark, Finland, Netherlands, Belgium, Switzerland, Austria, Italy, Spain, Portugal. *New World* — United States, Canada, Australia, New Zealand. *Latin America* — Argentina, Chile, Uruguay, Cuba, Costa Rica. *Other* — Sri Lanka (British Ceylon, active colonial education investment, known anomaly), Hong Kong.

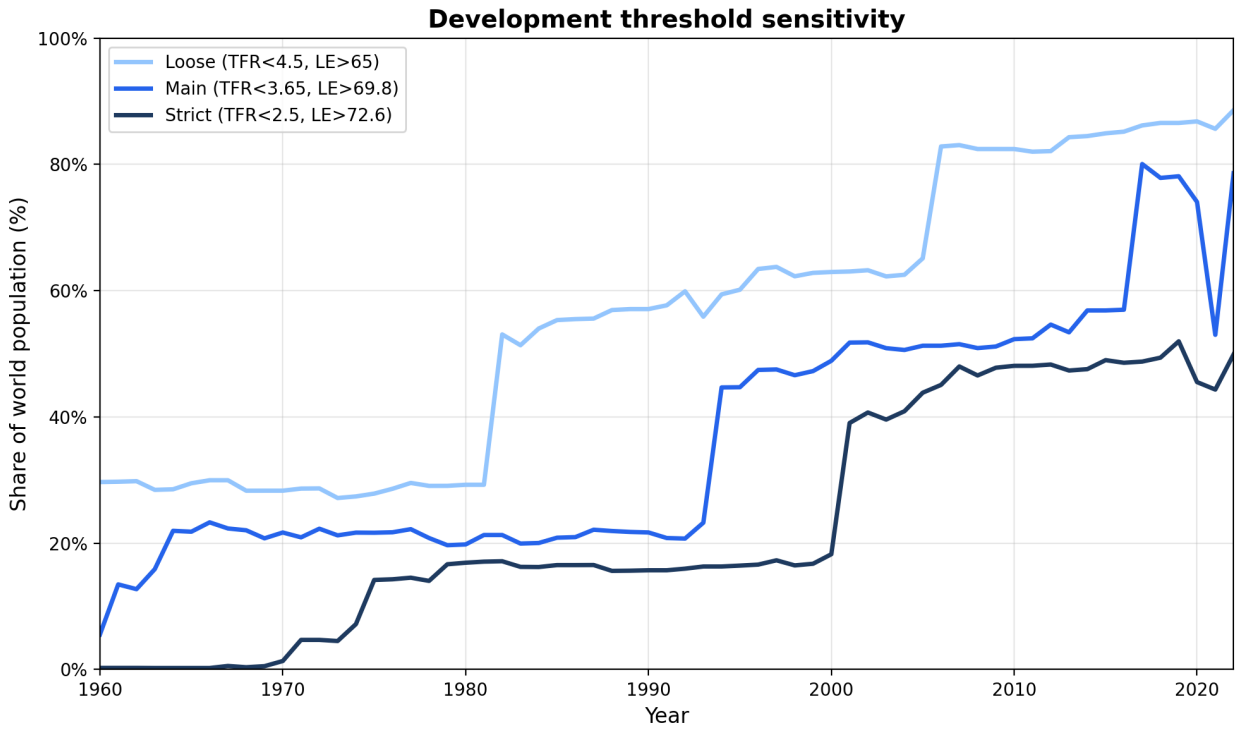

Figure 7: Cumulative share of world population crossing development thresholds under three specifications: loose (TFR < 4.5, LE > 65), main (TFR < 3.65, LE > 69.8), and strict (TFR < 2.5, LE > 72.6). The shape is identical across specifications — the same China jump, the same steady rise. The threshold choice shifts the level; it does not change the story. *(Source: World Bank WDI, WCDE v3. See* `scripts/figures/cumulative_developed_thresholds.py`*)*

Data source robustness

The paper uses WCDE v3, which reconstructs historical education distributions from census microdata, DHS surveys, and demographic back-projection. Three tests verify that the main forward-prediction and residualization results are not artifacts of WCDE's reconstruction methodology: replication on the independent Barro-Lee v3.0 dataset (Lee & Lee 2016) with different source surveys and interpolation; residualization on Barro-Lee; and restriction of WCDE to post-1970 observations, where data quality is highest and reconstruction plays the least role.

Specification	LE R^2	TFR R^2
WCDE v3 (main)	0.335	0.388
Barro-Lee v3.0 ("at least some secondary," age 15–24, 146 countries)	0.320	0.287
Barro-Lee v3.0 (mean years of schooling)	0.400	0.428
WCDE v3, post-1970 only	0.313	0.281
Barro-Lee residualization (GDP with education's contribution removed)	≤ 0.003	≤ 0.003

All specifications significant at $p < 0.0001$, same sign and direction. The finding that income has no independent predictive power is not specific to one dataset; the pre-1970 decades add statistical power but are not driving the result. The natural experiments (Cambodia, Bihar–Kerala, Asian financial crisis, Russia, South Africa) rest on WDI outcome data and documented historical events, not on WCDE reconstruction, and are unaffected by any data-source concern.[19]

B. Engagement with Alternative Frameworks

The main text presents the positive case. This appendix tests it against the major competing accounts. For orientation, here is the full set of frameworks the paper engages. Each identifies something real about development; each misses the educated substrate that produces what it identifies.

[19] See `scripts/robustness/barro_lee_replication.py`. Barro-Lee v3.0 source: `github.com/barrolee/BarroLeeDataSet`.

Smith (1776), division of labour. Specialisation → productivity → wealth. Smith took literate Scottish labour as given; the causal chain starts one link earlier, at categorical literacy. Main text: Section 1 and Section 4.3.

Sen, capabilities. Development is what people can do and be, not income. The capabilities are parallel expressions of educated cognition rather than a list of coequal axes. Main text: Section 3.

Drèze & Sen (1989), *Hunger and Public Action*. Democracy prevents famine; development runs through growth-mediated or support-led pathways. Bihar–Kerala 1966 isolates education within the same democracy (Section 4.10); China matched on mean years of schooling shows the support-led residual was education (Section 5.4).

Acemoglu & Robinson (2001, 2012). Settler mortality instruments for institutions. The instrument identifies Protestant schools, not institutions: first-stage F = 12.2 for education, 2.6 for institutions (Section B.2).

Deaton (2013), *The Great Escape*. Post-1950 mortality gains are health technology diffusion independent of education. Technology converts to outcomes only through educated households; residualized GDP R^2 never exceeds 0.023 (Section B.3).

The last three are developed in the subsections below.

B.1 The Institutional Challenge

Institutions are built by educated people. Controlling for institutions is controlling for a product of education.

Acemoglu & Robinson (2012) argue that institutions — inclusive versus extractive — are the primary determinant of long-run development.

Country fixed effects absorb time-invariant institutional characteristics. Time-varying institutional change is itself driven by education. Every measure of "institutional quality" is produced by educated populations. The civil servants, judges, journalists, health workers, and administrators who constitute institutional capacity are the educated cohort. Educated populations build institutions, not the reverse.

Qatar ($69,000 per capita, stable institutions by standard measures) delivered 4.8 percentage points below its generational education baseline in 2015. The institutions were present; the educational commitment was not; the development did not compound. The Philippines (Section 5.1) confirms the same point from the opposite direction.

India has had democratic institutions, an independent judiciary, and a free press since 1947. China had none. China expanded lower secondary completion from 10% to 75% in 40 years (1950–1990) at 1.6 pp/yr — under Mao, the Cultural Revolution, and the Great Leap Forward. India expanded from 10% to 37% in the same 40 years at 0.7 pp/yr — with every institutional advantage the literature claims to matter. India's institutions did not accelerate education; China's absence of them did not prevent it. Among countries still expanding, mean expansion rates were 1.06 pp/yr in 1950–1975, 0.86 in 1975–2000, and 0.94 in 2000–2015. If institutions drove educational expansion, rates should have risen as institutions matured. They did not.

Autocracy does not predict education speed. Merging Polity5 time-varying regime scores with the WCDE completion data (160 countries, 1950–2020, lagged 20 years to match the regime that made the schooling decision), polity2 explains less than 0.7% of the variance in education gain rates (Table A9). Democracies are slightly faster on average: 10.3 pp/decade against 8.1 for autocracies at the 20-year lag.

What autocracy produces is not speed but *variance*: 76% of autocratic countries fall below the democratic median gain rate, but the fat right tail includes Korea and Cuba. Autocracies populate both the fastest and the slowest education trajectories in the dataset. Among 50 countries that transitioned, 30 were faster under democracy (paired mean $= +0.6$ pp/decade, $p = 0.57$).

Regime type is a variance amplifier, not a cause. The decision variable is education policy (Section 6), and it is orthogonal to regime type. See `scripts/robustness/regime_education.py`.

B.2 The Colonial Test

The famous settler mortality instrument does not measure institutions. It measures whether Protestant colonisers — who built schools — survived.

Acemoglu, Johnson & Robinson (2001) use settler mortality as an instrument for institutional quality: where European settlers survived, they built inclusive institutions; where they did not, they built extractive ones. The instrument cannot distinguish this from an alternative channel: where *Protestant* settlers survived, they built *schools*. The Reformation made mass literacy a theological obligation (Section 2.2). The Counter-Reformation removed it. Protestant colonisers (Britain, the Netherlands) transplanted mass education systems. Catholic colonisers (Spain, Portugal, France, Belgium) did not. AJR's instrument captures this difference and attributes it to institutions.

The data separates the two channels. Among 99 former colonies, education at independence (1950 lower secondary completion) explains 47% of the variance in current GDP per capita. Polity5 institutional quality explains 11%. Colonial-era education (1900 birth cohort, people schooled under colonial rule) explains 38%. Coloniser religion alone explains 7% — but adding religion to a model that already contains education raises R^2 from 0.465 to 0.466. Religion predicts GDP only because it predicts education. Once education is in the model, religion adds nothing. The channel is religion → schools → education → development, not religion → institutions → development (Table A11).

A formal instrumental-variable test sharpens the result. The strategy uses AJR's own settler-mortality channel and asks what it actually identifies — not whether the instrument works, but whether its exogenous variation enters development through institutions or through schools. Two-stage least squares using coloniser religion as the excluded instrument produces a first-stage F-statistic of 12.2 for education at independence — above the Stock & Yogo (2005) threshold for instrument relevance — but only 2.6 for Polity2 institutional quality, well below the threshold (Table A13). The second stage for education is significant ($\hat{\beta} = +0.059$, $t = 3.21$, $p = 0.002$); the institutional 2SLS is unreliable because the instrument is weak. The Wu-Hausman test is insignificant ($p = 0.85$), confirming that OLS and 2SLS agree when the endogenous variable is education — consistent with coloniser religion being exogenous to the education channel. AJR's instrument works, but it works *because* it predicts education, not because it predicts institutions. See `scripts/robustness/iv_2sls_colonial.py`.

Latin America is the critical case. Spanish and Portuguese colonies were set-

tler colonial — Europeans came and stayed in large numbers, built cities, established legal systems. AJR's framework predicts that settlement should produce inclusive institutions and therefore development. It did not. AJR require an ad hoc distinction between "inclusive" and "extractive" settler colonialism to explain why Latin America underperforms. The education account needs no such distinction: Catholic settlers brought no mass education tradition. Spain had 0.6% primary completion for its own 1875 birth cohort; Portugal had 0.1%. They could not transplant what they did not possess. The educational base at independence was correspondingly low (mean 11% lower secondary, 1950), and development followed the education, not the institutions.

The settler mortality instrument is a Protestant education instrument. See `scripts/robustness/colonial_vs_institutions.py`.

The Protestant channel connects to Easterlin's (1981) original insight: the divergence in economic growth between early and late industrializers tracked the divergence in mass schooling — driven originally by the Reformation's insistence on personal scripture-reading, which produced mass literacy across Northern Europe. The distribution of development in 1960 was substantially the distribution of schooling in 1860. Goldin & Katz (2008) document the same transmission within the United States: the "human capital century" of 1910–1940, driven by the high school movement, explains the majority of twentieth-century income growth — a within-country longitudinal result consistent with the cross-country generational mechanism estimated here.

B.3 The Deaton Objection

Health technology has diffused globally. Outcomes have not. The difference is education.

Deaton (2013) attributes the curve's upward shift to global health technology diffusion independent of education. Vaccines, antibiotics, and oral rehydration protocols are cheap, well-understood, and globally available — the technology has diffused. But a vaccine in a warehouse is not a vaccinated child. Between the existence of a technology and its uptake sits every household decision this paper describes. The technology is globally available; the capacity to convert it into behaviour is locally determined by education. Diffusion without uniform outcomes is itself evidence that education is the operative

variable.

Table A9. Regime type and education gain rate. Polity5 time-varying regime scores (polity2, -10 to $+10$) merged with WCDE lower secondary completion (both sexes, age 20–24). Education at age 20–24 reflects schooling 15–20 years earlier; lagged specifications attribute the education outcome to the regime that made the decision. 160 countries, 1950–2020. See `scripts/robustness/regime_education.py`.

Regime lag	R^2(polity2)	Polity coef	Auto mean	Demo mean	n intervals
Concurrent (0 yr)	0.0015	+0.056	8.5	9.5	1,782
15-year lag	0.0067	+0.122	8.1	10.0	1,574
20-year lag	0.0050	+0.106	8.1	10.3	1,472

Gain rate in percentage points per decade. Polity coef: the slope of the democracy score on gain rate (positive = more democratic regimes gain faster). R^2: share of gain-rate variation explained by democracy score alone — less than 1% at every lag. Auto = strong autocracies (polity2 ≤ -6); Demo = strong democracies (polity2 $\geq +6$). Both invest at roughly the same rate.

Table A11. Colonial education vs. institutions as predictors of current GDP. 99 former colonies classified by coloniser identity and religion (Protestant: Britain, Netherlands; Catholic: Spain, Portugal, France, Belgium, Italy). Education data from WCDE v3; GDP from World Bank WDI (constant 2017 USD). See `scripts/robustness/colonial_vs_institutions.py`.

Predictor	R^2 (log GDP 2020)
Current education (2020 lower sec)	0.696
Education at independence (1950 lower sec)	0.465
Colonial-era education (1900 cohort primary)	0.377
Polity2 2015 (institutional quality)	0.114
Coloniser religion (Protestant = 1)	0.069
Education 1950 + coloniser religion	0.466

Each row is a separate bivariate OLS regression on log GDP per capita 2020, except the final row which includes both regressors. Adding coloniser religion

to education raises R^2 from 0.465 to 0.466: religion predicts GDP only through its effect on education.

Table A13. 2SLS instrumental-variable contest: education vs. institutions. Coloniser religion (Protestant = 1) as the excluded instrument. First-stage F > 10: strong instrument (Stock & Yogo 2005). 84 former colonies with complete data. See `scripts/robustness/iv_2sls_colonial.py`.

	First stage		Second stage (log GDP)	
Endogenous variable	F-stat	Strength	$\hat{\beta}$	t
Education 1950 (lower sec)	12.2	Strong	+0.059	3.21
Polity2 (institutions)	2.6	Weak	+0.349	1.56†

†*Institutional coefficient is inflated by weak-instrument bias ($F = 2.6 < 10$); the 2SLS estimate is unreliable. Wu-Hausman test for the education specification: $p = 0.85$ (OLS and 2SLS agree). The instrument predicts education, not institutions.*